Quantum Complexity Theory

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Part III

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Chapter 8

Quantum Merlin–Arthur

The abstract notion of *proofs* — particularly efficiently verifiable proofs — is central to complexity theory. For instance, the complexity class NP is representative of this notion: a polynomial-length string (namely, a witness) plays the role of a purported proof that an input string is a yes-input to some promise problem, where the validity of the proof is effectively checked by a polynomial-time deterministic computation. There are many known extensions of this concept, including so-called Arthur–Merlin games and interactive proof systems.

In this chapter we will define a quantum analogue of NP, known as QMA, where a quantum state plays the role of a proof, which is checked for validity by an efficient quantum computation. In fact, it is more accurate to say that QMA is a quantum analogue of the complexity class MA (short for *Merlin–Arthur*), which is similar to NP except that the proof verification is a probabilistic computation rather than a deterministic one.

We will discuss QMA, as well as some variations on it, for the next several chapters. In particular, we will examine a couple of different methods for error reduction for QMA, prove the containment QMA $\subseteq$ PP, and establish the completeness of a promise problem called the *local Hamiltonian* problem for QMA.

For this first chapter on QMA, in addition to defining the class, we will examine some basic properties of it, and discuss two examples of problems in QMA. One is effectively a canonical complete promise problem for the class and the other is a language connected with finite groups known as the *group non-membership* problem.

8.1 Merlin–Arthur and Arthur–Merlin

We will begin the chapter by discussing two classical complexity classes, MA and AM, which are short for *Merlin–Arthur* and *Arthur–Merlin*. These classes, along with their Camelot-inspired nomenclature, were introduced by László Babai in 1985.

The idea is that Arthur is a wise king, but as a human lacking mystical powers he is limited to performing polynomial-time computations. He does have a coin to flip, though, so his computations can make use of randomness. Merlin, on the other hand, is a cambion wizard — born of a human mother and demon father — possessing supernatural powers as a result. He can perform arbitrary computations at unit cost, but he is not trustworthy and will lie about his computations to serve his own interests. Arthur–Merlin games represent interactions between these two characters, in which Arthur attempts to tap into Merlin's power while taking responsibility for the correctness of his own decisions.

The names of the classes MA and AM reflect the order in which the two characters act in a hypothetical interaction: either Merlin acts first and Arthur acts second, or the other way around. Additional classes can be defined by allowing for more than one action each, but we will delay any discussion of this possibility for now and revisit it in the context of interactive proof systems. Precise definitions of the classes MA and AM appear below.

Two notational conventions from Chapter 2, which will be used throughout this chapter, are worth recalling at this point. First, whenever an input string x has been fixed or is implicit from the context, we write $n = |x|$ and write p in place of $p(n)$ or $p(|x|)$ to refer to the value of a polynomial resource bound p at n. Second, recall the Я quantifier, read as "for most," where $\big(\text{Я}\, z \in \{0,1\}^m\big)\big[P(z)\big]$ means that a predicate P holds for at least two thirds of all strings z of length m. The quantifier Я^c is defined similarly but with the threshold $2/3$ replaced by c.

The class MA

Now we are ready to define MA, the first of the two Arthur–Merlin classes to be introduced in this chapter. In this case, Merlin goes first, presenting Arthur with a polynomial-length witness as in the definition of NP, and Arthur goes second, generating a sequence of random bits that he uses when verifying Merlin's witness.

Definition 8.1. A promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ over an alphabet Σ is contained in MA if there exist polynomial resource bounds p and q and a language

$B \in \mathrm{P}$ such that these implications are true.

$$x \in A_{\text{yes}} \implies \left(\exists\, y \in \{0,1\}^p\right)\left(\text{Я}\, z \in \{0,1\}^q\right)\left[\langle x,y,z\rangle \in B\right] \tag{8.1}$$

$$x \in A_{\text{no}} \implies \left(\forall\, y \in \{0,1\}^p\right)\left(\text{Я}\, z \in \{0,1\}^q\right)\left[\langle x,y,z\rangle \notin B\right] \tag{8.2}$$

Note that the ordering of the quantifiers in this definition is critically important: Merlin selects his witness first and Arthur chooses his random bits second. Switching the order gives us the complexity class AM, to be discussed in the next subsection.

The following two containments follow immediately from the definition, taking either $p = 0$ or $q = 0$, respectively.

Proposition 8.2. $\mathrm{BPP} \subseteq \mathrm{MA}$.

Proposition 8.3. $\mathrm{NP} \subseteq \mathrm{MA}$.

Let us also observe that the constants $2/3$ and $1/3$, which appear in the definition of the Я quantifier, can be modified without changing the class that is defined. In essence, the same range of choices that define BPP can be used here to define MA. The following proposition reflects this freedom for exponentially small error bounds.

Proposition 8.4 (Error reduction for MA)**.** *Let $A = (A_{\text{yes}}, A_{\text{no}})$ be a promise problem contained in* MA*, taking p and q to be polynomial resource bounds and $B \in \mathrm{P}$ to be a language such that the conditions of Definition 8.1 are satisfied. For every polynomial resource bound r, which defines an error bound $\varepsilon = \varepsilon(n) = 2^{-r(n)}$, there exist a polynomial resource bound s and a language $C \in \mathrm{P}$ such that*

$$x \in A_{\text{yes}} \implies \left(\exists\, y \in \{0,1\}^p\right)\left(\text{Я}^{1-\varepsilon}\, z \in \{0,1\}^s\right)\left[\langle x,y,z\rangle \in C\right], \tag{8.3}$$

$$x \in A_{\text{no}} \implies \left(\forall\, y \in \{0,1\}^p\right)\left(\text{Я}^{1-\varepsilon}\, z \in \{0,1\}^s\right)\left[\langle x,y,z\rangle \notin C\right]. \tag{8.4}$$

Remark 8.5. Notice that the witness length p in (8.3) and (8.4) is unchanged. The fact that the error ε can be taken to be smaller than the reciprocal of 2^p, the total number of witnesses, is particularly important and has interesting implications.

Conceptually speaking, the proof is routine: running several independent verifications, using different random bits for z each time, followed by a majority vote, decreases error exponentially in the number of independent runs. The analysis is similar to the one already given for BPP in Chapter 2.

The class AM

The second class in the Arthur–Merlin framework we will consider is AM, where Arthur acts first by generating a random bit string. Merlin then responds by presenting Arthur with a witness, which now may depend on Arthur's random bits. The final step, which is that Arthur performs a deterministic polynomial-time computation, is the same as before — this is always the final step for Arthur–Merlin games, and is not considered as a turn for Arthur. Here is a formal definition of this class.

Definition 8.6. A promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ over an alphabet Σ is contained in AM if there exist polynomial resource bounds p and q and a language $B \in \mathrm{P}$ such that these implications are true.

$$x \in A_{\text{yes}} \implies \big(\text{ꟻ}\, z \in \{0,1\}^q\big)\big(\exists\, y \in \{0,1\}^p\big)\big[\langle x,y,z\rangle \in B\big] \tag{8.5}$$

$$x \in A_{\text{no}} \implies \big(\text{ꟻ}\, z \in \{0,1\}^q\big)\big(\forall\, y \in \{0,1\}^p\big)\big[\langle x,y,z\rangle \notin B\big] \tag{8.6}$$

Thus, the definitions for MA and AM differ only in the ordering of the quantifiers. This change in order has a significant effect: MA $\subseteq$ AM whereas the reverse containment is not known to be true. Several interesting and natural problems are in AM but not known to be in MA, including the *graph non-isomorphism* problem. For this problem, the yes-inputs are pairs of (simple and undirected) graphs that are *not* isomorphic, while the no-inputs are pairs of isomorphic graphs. On the other hand, there are plausible Boolean circuit lower bounds that would, if true, imply the collapse NP $=$ MA $=$ AM.

Proposition 8.7. MA $\subseteq$ AM.

Proof. Let $A = (A_{\text{yes}}, A_{\text{no}})$ be contained in MA. By the definition of MA together with Proposition 8.4, there exist polynomial resource bounds p and s together with a language $C \in \mathrm{P}$ such that

$$x \in A_{\text{yes}} \implies \big(\exists\, y \in \{0,1\}^p\big)\big(\text{ꟻ}^{1-\varepsilon}\, z \in \{0,1\}^s\big)\big[\langle x,y,z\rangle \in C\big], \tag{8.7}$$

$$x \in A_{\text{no}} \implies \big(\forall\, y \in \{0,1\}^p\big)\big(\text{ꟻ}^{1-\varepsilon}\, z \in \{0,1\}^s\big)\big[\langle x,y,z\rangle \notin C\big], \tag{8.8}$$

for $\varepsilon = \frac{1}{4}\cdot 2^{-p}$. Notice in particular that we are taking the error to be smaller than the reciprocal of the total number of witness strings $y \in \{0,1\}^p$, which is critical to the proof.

The implication

$$\begin{aligned}&\bigl(\exists\, y \in \{0,1\}^p\bigr)\bigl(\mathcal{R}^{1-\varepsilon}\, z \in \{0,1\}^s\bigr)\bigl[\langle x,y,z\rangle \in C\bigr]\\ &\qquad\Longrightarrow\ \bigl(\mathcal{R}^{1-\varepsilon}\, z \in \{0,1\}^s\bigr)\bigl(\exists\, y \in \{0,1\}^p\bigr)\bigl[\langle x,y,z\rangle \in C\bigr]\end{aligned} \tag{8.9}$$

is immediate, as the left-hand side implies that the same choice of $y \in \{0,1\}^p$ can be made on the right-hand side for every random bit string $z \in \{0,1\}^s$, with no dependence on z required. Hence

$$x \in A_{\text{yes}} \implies \bigl(\mathcal{R}^{1-\varepsilon}\, z \in \{0,1\}^s\bigr)\bigl(\exists\, y \in \{0,1\}^p\bigr)\bigl[\langle x,y,z\rangle \in C\bigr]. \tag{8.10}$$

Now suppose that $x \in A_{\text{no}}$, and for each $y \in \{0,1\}^p$ let

$$Z_y = \bigl\{z \in \{0,1\}^s \,:\, \langle x,y,z\rangle \in C\bigr\}, \tag{8.11}$$

which is the set of random bit strings that lead Arthur to a wrong answer (essentially representing false positives) on input x when Merlin's witness is y. By (8.8) we have $|Z_y| \leq \varepsilon\, 2^s$ for each $y \in \{0,1\}^p$, and therefore, by the union bound,

$$\left|\bigcup_{y\in\{0,1\}^p} Z_y\right| \leq \sum_{y\in\{0,1\}^p} |Z_y| \leq \frac{2^s}{4}. \tag{8.12}$$

There are therefore at least $\frac{3}{4}\cdot 2^s$ strings $z \in \{0,1\}^s$ for which *no* strings $y \in \{0,1\}^p$ satisfy $\langle x,y,z\rangle \in C$. That is,

$$x \in A_{\text{no}} \implies \bigl(\mathcal{R}\, z \in \{0,1\}^s\bigr)\bigl(\forall\, y \in \{0,1\}^p\bigr)\bigl[\langle x,y,z\rangle \notin C\bigr]. \tag{8.13}$$

By (8.10) and (8.13) it follows that $A \in \mathrm{AM}$. □

Finally, we observe the following simple containment. (Better containments are known, including the containment of AM in the second level of the so-called polynomial-time hierarchy — but having not discussed this hierarchy yet, we will put this off for now.)

Proposition 8.8. $\mathrm{AM} \subseteq \mathrm{PSPACE}$.

Proof. Let $A = (A_{\text{yes}}, A_{\text{no}})$ be contained in AM, and let p, q, and B be as in Definition 8.6. Consider a computation that inputs x, initializes a counter to 0, and iterates through all $z \in \{0,1\}^q$, searching for a string $y \in \{0,1\}^p$ that satisfies $\langle x,y,z\rangle \in B$. If such a string y is found, the counter is incremented and the next z is considered. Once all z have been considered, the input is accepted if the counter is at least $\frac{2}{3}\cdot 2^q$, and rejected otherwise. This computation decides A and, by reusing the same space to check if $\langle x,y,z\rangle \in B$ for every y and z, can be performed in polynomial space. □

8.2 Quantum Merlin–Arthur

The class QMA

We now define QMA, which represents a natural quantum analogue of the complexity class MA. Specifically, Merlin's witness now takes the form of a quantum state rather than a classical string, while Arthur's selection of a random bit string together with his verification are replaced by a polynomial-time quantum computation.

Definition 8.9. Let $A = (A_{\text{yes}}, A_{\text{no}})$ be a promise problem over an alphabet Σ, let $\alpha, \beta : \mathbb{N} \to [0,1]$ be functions (or constants), and let p be a polynomial resource bound. Then $A \in \text{QMA}_p(\alpha, \beta)$ if there exists a polynomial-time uniform family $\{Q_x : x \in \Sigma^*\}$ of quantum circuits, where each Q_x has p input qubits and one output qubit, such that these implications are true.

$$x \in A_{\text{yes}} \implies \langle 1| Q_x(\rho) |1\rangle \geq \alpha(|x|) \text{ for some state } \rho \text{ of } p \text{ qubits} \tag{8.14}$$

$$x \in A_{\text{no}} \implies \langle 1| Q_x(\rho) |1\rangle \leq \beta(|x|) \text{ for every state } \rho \text{ of } p \text{ qubits} \tag{8.15}$$

We write $\text{QMA}(\alpha, \beta)$ for the union of $\text{QMA}_p(\alpha, \beta)$ over all polynomial resource bounds p, and the complexity class QMA is defined as $\text{QMA} = \text{QMA}(2/3, 1/3)$.

The circuits Q_x appearing in this definition will be referred to as *Arthur's verification circuits*: on input x, Arthur runs Q_x on the witness state that Merlin provides, accepting or rejecting according to the output qubit. For each p-qubit state ρ, the probabilities that Q_x accepts and rejects the witness ρ are as follows.

$$\Pr(Q_x \text{ accepts } \rho) = \langle 1|Q_x(\rho)|1\rangle \tag{8.16}$$

$$\Pr(Q_x \text{ rejects } \rho) = \langle 0|Q_x(\rho)|0\rangle \tag{8.17}$$

We note that the state ρ, which represents Merlin's quantum witness, is taken to be a density matrix in this definition — but it could be replaced by a pure state $|\psi\rangle$ without changing the class.

Containments among MA, AM, and QMA

By taking $p = 0$ the following (rather obvious) containment is obtained.

Proposition 8.10. $\text{BQP} \subseteq \text{QMA}$.

It is also the case that MA $\subseteq$ QMA — and though this containment is conceptually straightforward, it is perhaps not quite as immediate as the previous one.

Proposition 8.11. MA $\subseteq$ QMA.

This proposition follows from a few basic facts: 1) Polynomial-time deterministic computations can be simulated by polynomial-time generated families of Boolean circuits (which can, in turn, be simulated by quantum circuits). 2) Random bits can be generated by quantum circuits. 3) We can measure (or dephase) Merlin's quantum witness with quantum circuits to obtain a convex combination of classical witnesses.[1] A straightforward analysis combining these facts proves the proposition.

It is not known that AM $\subseteq$ QMA, and indeed AM $\subseteq$ PP is not known to be true. Both inclusions would follow trivially from AM $=$ NP, which is plausible as was already mentioned. On the other hand, a 1992 result of Vereshchagin provides an oracle A relative to which $\mathrm{AM}^A \not\subseteq \mathrm{PP}^A$, which limits the proof techniques through which we might otherwise hope to prove this containment. We will discuss oracle results like this and how they can be interpreted later in the course.

Let us also take note of the following theorem, whose proof will be covered in Chapter 9. It makes use of the same GapP machinery that we used to prove BQP $\subseteq$ PP, but it is greatly simplified through error reduction for QMA, particularly through a method that maintains the number of qubits of Merlin's witness state. A key aspect of the proof is that the error is reduced below the reciprocal of the dimension of the witness space.

Theorem 8.12. QMA $\subseteq$ PP.

Figure 8.1 illustrates how MA, AM, and QMA relate to the other complexity classes we have seen thus far in the course.

8.3 Two problems in QMA

To conclude the chapter, we will discuss two examples of problems in QMA. The first is a canonical QMA-complete promise problem that is analogous to the circuit satisfiability problem for Boolean circuits. The second problem is called the group non-membership problem, and is particularly interesting because it can be described as a language (i.e., a promise problem for which the promise is trivial).

[1] Doing this is not actually necessary, but arguably makes the proof more transparent.

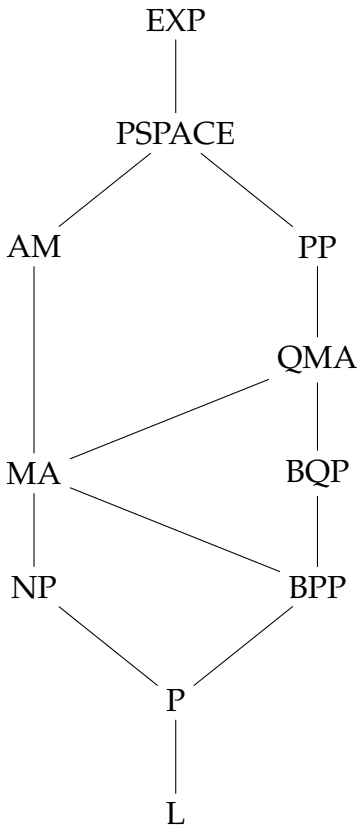


Figure 8.1: A Hasse diagram extending Figure 6.1 to include MA, AM, and QMA.

Quantum circuit satisfiability

As suggested above, our first example of a problem in QMA is a quantum analogue of the circuit satisfiability problem from Chapter 3.

QUANTUM CIRCUIT SATISFIABILITY (QCIRCUIT-SAT)

Input	A quantum circuit Q having m input qubits and one output qubit, for some $m \in \mathbb{N}$.
Yes	There exists a state ρ of m qubits for which $\langle 1\vert Q(\rho)\vert 1\rangle \geq 2/3$.
No	Every state ρ of m qubits satisfies $\langle 1\vert Q(\rho)\vert 1\rangle \leq 1/3$.

Proposition 8.13. QCIRCUIT-SAT *is* QMA*-complete.*

Proof. First let us verify that QCIRCUIT-SAT $\in$ QMA. Definition 8.9 demands that every circuit Q_x in the family have $p = p(n)$ input qubits, where $n = |x|$ as usual,

for some polynomial resource bound p. We will take $p(n) = n$, which will be sufficient, as we will see. The specific way that each Q_x is defined is as follows.

If x encodes a quantum circuit Q having m input qubits and one output qubit, then Q_x is taken to be this same circuit, except with enough additional input qubits so that the total number of input qubits is $n = |x|$. (Note that n must exceed m, as the length of the encoding of any circuit exceeds its number of input qubits.) In addition, erasure gates are applied to the additional input qubits that are not used by Q. It is the case that

$$\max_{\rho} \langle 1|Q_x(\rho)|1\rangle = \max_{\sigma} \langle 1|Q(\sigma)|1\rangle, \tag{8.18}$$

where the first maximum is over all states ρ of n qubits and the second is over all states σ of m qubits. This is because every ρ determines a unique σ through partial tracing over $n - m$ qubits and every σ can be obtained in this way (for infinitely many different states ρ).

For every other string x, the circuit Q_x consists of n erasure gates applied to n input qubits along with one ancillary gate that produces the output $|0\rangle$.

The family $\{Q_x\}$ is polynomial-time uniform, as each circuit is read directly from the input when it encodes a valid quantum circuit, and the added qubits and erasure gates are easily included in polynomial time. The yes- and no-conditions of the problem therefore establish the two implications in Definition 8.9, for $\alpha = 2/3$ and $\beta = 1/3$.

It remains to prove that QCIRCUIT-SAT is QMA-hard, which follows directly from the definition of QMA itself. Let $A = (A_{\text{yes}}, A_{\text{no}})$ be a promise problem contained in QMA, and take $\{R_x : x \in \Sigma^*\}$ to be a polynomial-time uniform family of quantum circuits satisfying the conditions of Definition 8.9 (using the name q rather than p for the polynomial resource bound, so that each R_x has q input qubits and one output qubit). Define a function $f : \Sigma^* \to \{0,1\}^*$ as

$$f(x) = \langle R_x \rangle \tag{8.19}$$

for every string $x \in \Sigma^*$. The fact that this function is computable in polynomial time is precisely the statement that the family $\{R_x\}$ is polynomial-time uniform: each encoding $\langle R_x \rangle$ is the output of the polynomial-time DTM that generates the family.

If $x \in A_{\text{yes}}$, then there exists a state ρ of q qubits for which $\langle 1|R_x(\rho)|1\rangle \geq 2/3$, so $\langle R_x \rangle$ is a yes-input to QCIRCUIT-SAT; and if $x \in A_{\text{no}}$, then every state ρ of q qubits satisfies $\langle 1|R_x(\rho)|1\rangle \leq 1/3$, so $\langle R_x \rangle$ is a no-input. Thus $A \leq_{\text{m}}^{\text{p}}$ QCIRCUIT-SAT. As A was an arbitrary problem contained in QMA, this establishes the QMA-hardness of QCIRCUIT-SAT. □

This is a rather generic way of defining a complete problem for a class, essentially turning the definition of the class into a promise problem without much thought or creativity. A structurally more interesting QMA-complete problem, known as the local Hamiltonian problem, is the topic of Chapter 10.

Group non-membership

Our second example of a problem in QMA has a different character and requires some set-up, which we will try to keep as simple as possible for the sake of this chapter.

We assume the existence of a family of underlying finite groups, together with a way of encoding group elements as binary strings having a fixed length for each group in the family. There is an important caveat, which is that every group element must have a *unique* encoding as a binary string. In addition, we require the existence of a mechanism to perform the group operations efficiently. This set-up can be made quite general through the notion of *group oracles*, but for the sake of this example it is reasonable to imagine that the underlying groups are the groups of invertible matrices over some finite field (where testing group membership is conjectured to be difficult). For whatever finite groups are in place, the assumption that the group operations can be performed efficiently is formalized as the existence of two polynomial-time uniform families of quantum circuits implementing the unitary operations

$$|g\rangle|h\rangle \mapsto |hg\rangle|h\rangle \qquad \text{and} \qquad |g\rangle \mapsto |g^{-1}\rangle \tag{8.20}$$

on group elements g and h (one family for each operation).

Assuming such a set-up is in place, the group non-membership problem is as follows.

GROUP NON-MEMBERSHIP (GNM)

Input	Group elements $g_1, \ldots, g_k$ and a.
Yes	$a \notin \langle g_1, \ldots, g_k \rangle$.
No	$a \in \langle g_1, \ldots, g_k \rangle$.

Here $\langle g_1, \ldots, g_k \rangle$ denotes the group generated by $g_1, \ldots, g_k$, which we will denote by G for the remainder of the chapter.

It might be tempting to say that "subgroup non-membership" is a more fitting name for this problem, as G is a subgroup of whatever underlying group we are working with, but it is a standard assumption in computational group theory that

groups are given by generators in this way. Taking this point of view, the group of invertible matrices over a finite field (or a hypothetical black-box group) really serves as a generic super-group in which we are working, not the group G that the problem itself concerns.

The complement of this problem, which is naturally named the *group membership* problem, is known to be in NP. This follows from the fact that, if $a \in \langle g_1, \ldots, g_k \rangle$, there must exist a so-called *straight-line program* having length polynomial in the length of group element encodings that prescribes a way of generating a from $g_1, \ldots, g_k$, which can be verified in polynomial time. Group non-membership, on the other hand, is not known to be in MA. (It is, however, in AM.)

The idea behind the proof that this problem is in QMA is to hypothesize that Merlin's witness is the *uniform superposition* over all of the elements of the input group $G = \langle g_1, \ldots, g_k \rangle$ (which, interestingly, does not depend on a at all). In general, for any nonempty finite set S (of group elements in this case), we write

$$|S\rangle = \frac{1}{\sqrt{|S|}} \sum_{g \in S} |g\rangle \tag{8.21}$$

to denote the uniform superposition over the elements of S. Thus, we hypothesize that Merlin's witness is $|G\rangle$, at least when he is being honest.

Let us also introduce the notation M_a to mean the unitary operation obtained by multiplying a given group element on the left by a, in place, so that

$$M_a |g\rangle = |ag\rangle. \tag{8.22}$$

This operation is obtained from the first operation in (8.20) by hard-coding $h = a$. If G is a finite group, and a is a group element, we have

$$a \in G \implies M_a |G\rangle = |aG\rangle = |G\rangle \tag{8.23}$$

$$a \notin G \implies M_a |G\rangle = |aG\rangle \perp |G\rangle. \tag{8.24}$$

Given the state $|G\rangle$ along with a group element a, it is straightforward to check whether or not $a \in G$ using the circuit in Figure 8.2, with $|\psi\rangle = |G\rangle$ and $U = M_a$. In general, the measurement outcome is 0 with probability

$$\frac{1}{4} \big\| |\psi\rangle + U|\psi\rangle \big\|^2 = \frac{1 + \Re\langle \psi | U | \psi \rangle}{2}, \tag{8.25}$$

and conditioned on that outcome, the state of the lower register becomes

$$\frac{|\psi\rangle + U|\psi\rangle}{\big\| |\psi\rangle + U|\psi\rangle \big\|}. \tag{8.26}$$

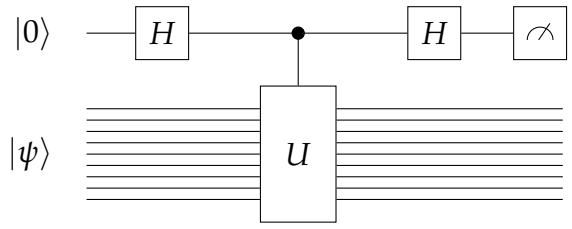


Figure 8.2: The controlled-unitary test, applied to a state $|\psi\rangle$ and a unitary operation U. The outcome of the measurement is 0 with probability $\big(1 + \Re\langle\psi|U|\psi\rangle\big)/2$.

For $|\psi\rangle = |G\rangle$ and $U = M_a$ we have

$$a \in G \implies \Pr(\text{output is } 0) = 1, \tag{8.27}$$

$$a \notin G \implies \Pr(\text{output is } 0) = 1/2. \tag{8.28}$$

So, if Merlin truly gives Arthur $|G\rangle$, he can use this state to check membership in G efficiently. (The error is $1/2$ when $a \notin G$, but this can be reduced as discussed in Chapter 9.)

Of course, the problem is that Merlin cannot be trusted and might present Arthur with a different state. For example, if Merlin instead gives Arthur $|g\rangle$, for some arbitrary group element g, Arthur will be tricked into believing $a \notin G$, regardless of whether this is true or not. The remedy is for Arthur to perform a test on the state he receives from Merlin prior to using it to check if $a \notin G$. The test will either fail, in which case Arthur learns that Merlin is being dishonest and rejects, or it passes. If Arthur's test passes, Merlin's witness may change as a result, but it will actually become better in this case, allowing Arthur to check if $a \notin G$ more reliably.

A technical fact is required to proceed, which we will treat as a black box for the sake of this chapter. First, we have the following definition.

Definition 8.14. Let G be a finite group and let $\varepsilon > 0$. A sequence $h_1, \ldots, h_N$ of elements of G is an *ε-uniform Erdős–Rényi generating sequence* for G if the distribution of the group element

$$h_1^{c_1} \cdots h_N^{c_N} \tag{8.29}$$

for $(c_1, \ldots, c_N) \in \{0,1\}^N$ chosen uniformly at random, is within ε of the uniform distribution over G in total variation distance.[2]

[2]The total variation distance between two probability distributions μ and ν over a finite set S is

Such sequences always exist and can be found efficiently, as the following theorem of Babai establishes.[3]

Theorem 8.15 (Babai). *There is a randomized polynomial-time procedure that, given generators $g_1, \ldots, g_k$ of a group G and a positive rational number ε, outputs a sequence $h_1, \ldots, h_N$ of elements of G that is an ε-uniform Erdős–Rényi generating sequence for G with probability at least $1 - \varepsilon$.*

We now have the components required. Arthur's test, together with the final check involving a, is as follows. Here we set $\varepsilon = 1/8$ and let W denote the q-qubit register containing Merlin's witness, where q is the encoding length of group elements.

1. Randomly select $h_1, \ldots, h_N \in G$ using the procedure of Theorem 8.15, so that with probability at least $1 - \varepsilon$ this sequence is an ε-uniform Erdős–Rényi generating sequence for $G = \langle g_1, \ldots, g_k \rangle$.
2. For each $j = N, N-1, \ldots, 1$, in this order, perform the controlled-unitary test of Figure 8.2 on the state contained in W and the unitary operation M_{h_j}. Reject if any of these tests produces the outcome 1.
3. Perform the controlled-unitary test on the state contained in W and the unitary operation M_a. Accept if the outcome is 1, and reject if the outcome is 0.

To see why this works, assume without loss of generality that Merlin presents Arthur with a pure state $|\psi\rangle$. (It suffices to consider only pure-state witnesses because the acceptance probability of any mixed state is a convex combination of acceptance probabilities for pure states.) Then, conditioned on all of the tests in step 2 producing the outcome 0, the state left in the register W is given by

$$\frac{T|\psi\rangle}{\|T|\psi\rangle\|} \tag{8.30}$$

where

$$T = \frac{1}{2^N}\bigl(\mathbb{1} + M_{h_1}\bigr) \cdots \bigl(\mathbb{1} + M_{h_N}\bigr). \tag{8.31}$$

defined as $\frac{1}{2}\sum_{s \in S} |\mu(s) - \nu(s)|$. Intuitively speaking, if one of μ or ν is chosen, each with probability $1/2$, and a single sample is drawn from the chosen distribution, then the optimal probability of correctly guessing which distribution produced the sample is $\frac{1}{2} + \frac{1}{2}\delta$, where δ is the total variation distance between μ and ν.

[3] Babai in fact proves a stronger statement, but the one given here is sufficient for our needs.

We do not actually need to normalize like this, however, as we are interested in the probability of accepting in step 3; this probability is given by

$$\Pr(\text{the procedure accepts } |\psi\rangle) = \frac{1}{4}\big\|(\mathbb{1} - M_a)\, T|\psi\rangle\big\|^2. \tag{8.32}$$

Now, if $h_1, \ldots, h_N$ is indeed an ε-uniform Erdős–Rényi generating sequence for $G = \langle g_1, \ldots, g_k \rangle$, then T must be close to the projection

$$\Pi_G = \frac{1}{|G|} \sum_{g \in G} M_g. \tag{8.33}$$

More precisely, using the fact that $M_g M_h = M_{gh}$ for all $g, h \in G$, expanding the product in (8.31) gives

$$T = \sum_{g \in G} \nu(g) M_g, \tag{8.34}$$

where ν is the distribution of the product $h_1^{c_1} \cdots h_N^{c_N}$ for $(c_1, \ldots, c_N) \in \{0,1\}^N$ chosen uniformly at random. Each M_g is unitary, so the triangle inequality gives

$$\|T - \Pi_G\| = \left\| \sum_{g \in G} \Big(\nu(g) - \frac{1}{|G|} \Big) M_g \right\| \leq \sum_{g \in G} \left| \nu(g) - \frac{1}{|G|} \right| \leq 2\varepsilon, \tag{8.35}$$

where the last inequality holds because the sum is twice the total variation distance between ν and the uniform distribution over G.

If $a \in G$, then $M_a \Pi_G = \Pi_G$, and together with (8.35) it follows that

$$\|(\mathbb{1} - M_a)\, T\| \leq 4\varepsilon. \tag{8.36}$$

Arthur therefore accepts with probability at most $4\varepsilon^2$, regardless of what state Merlin gives him. There is also the possibility that $h_1, \ldots, h_N$ is not an ε-uniform Erdős–Rényi generating sequence for $G = \langle g_1, \ldots, g_k \rangle$, but this happens with probability at most ε and we need not worry about what happens in this situation. The end result is that Arthur accepts when $a \in G$ with probability at most $\varepsilon + 4\varepsilon^2 < 1/4$.

On the other hand, if $a \notin G$, then the procedure still accepts the witness $|G\rangle$ with probability $1/2$. This is because each $h_j \in G$ and therefore $T|G\rangle = |G\rangle$; so each of the tests in step 2 passes, causing no change to the witness, leading to acceptance in step 3 with probability $1/2$ just like before.

It follows that GNM $\in$ QMA$(1/2, 1/4)$, and therefore GNM $\in$ QMA by error reduction, which is the subject of Chapter 9.

Chapter 9

Error reduction for QMA

In this chapter we will discuss error reduction for QMA. Error reduction for the class MA — where the witness is classical — is a simple matter of performing several independent verifications of a given witness, followed by a majority vote (or, more generally, a threshold computation). In contrast, error reduction for QMA is more subtle, due to the fact that quantum witnesses are generally disturbed by the process of verification. Thus, there is no guarantee that a given quantum witness will survive one verification intact to be available for the next. There are two main approaches to confronting this problem.

The first approach, called *parallel error reduction*, requires an increase in quantum witness size. In short, Merlin is asked to provide many copies of a given witness state, which Arthur verifies independently, followed by a majority vote or threshold computation to determine acceptance or rejection. Of course, Merlin is not trustworthy, so there is no guarantee that he will do what is expected. He could give Arthur a highly entangled state rather than independent copies of a witness, but an analysis reveals that this gives Merlin no advantage.

The second approach, called *in-place error reduction*, requires no increase in witness size. In fact, it requires no change at all to Merlin's witness, provided it is optimal for the original verification procedure. In this approach, Arthur's verification procedure is dilated to a unitary circuit — allowing it to be run in reverse — and successive verifications and reverse-verifications are interleaved with measurements that provide sufficient information to reduce error.

There are hypothetical situations in which parallel error reduction is the better option, such as one in which verifications must be performed by low-depth

quantum circuits. In-place error reduction, on the other hand, pays immediate dividends from a complexity-theoretic viewpoint, as it allows the error to be reduced below the reciprocal of the witness-space dimension. Much like we have already seen (in Corollary 2.17 and the proof of Proposition 8.7), this has interesting implications. Most notably, it leads to a simple proof that QMA $\subseteq$ PP, and allows us to conclude that logarithmic-length quantum witnesses provide no power beyond BQP. It will also be used in Chapter 10, in proving that the local Hamiltonian problem is QMA-hard.

9.1 Parallel error reduction

We begin with parallel error reduction for QMA. The method, which was mentioned briefly above, is described in more precise terms in Construction 9.1. In it we write $|y|_1$ for the Hamming weight of a binary string y, as in Chapter 2. In this construction, as well as in the proof of the theorem that follows, we take the input string x to be fixed, so that $Q = Q_x$, to avoid unnecessary notational clutter in the analysis.

Construction 9.1. Let Q be a quantum circuit taking a witness register of p qubits as input and producing one output qubit, let N be a positive integer, and let t be a real number representing a threshold value for acceptance. Define R to be the quantum circuit that takes witness registers $\mathsf{W}_1, \ldots, \mathsf{W}_N$, each of p qubits, and operates as follows.

1. Run Q independently on each of $\mathsf{W}_1, \ldots, \mathsf{W}_N$, letting $y = y_1 \cdots y_N \in \{0,1\}^N$ be the binary string whose bits record the outcomes.
2. Accept if $|y|_1 \geq t$, and reject otherwise.

Figure 9.1 illustrates this procedure in the form of a quantum circuit when $N = 5$.

Theorem 9.2. *Let $\alpha, \beta : \mathbb{N} \to [0,1]$ be polynomial-time computable functions for which there exists a polynomial resource bound s satisfying $\alpha(n) - \beta(n) \geq 1/s(n)$ for every n, and let p and r be polynomial resource bounds. Then*

$$\mathrm{QMA}_p(\alpha, \beta) \subseteq \mathrm{QMA}_q\bigl(1 - 2^{-r}, 2^{-r}\bigr) \tag{9.1}$$

for $q = 2prs^2$.

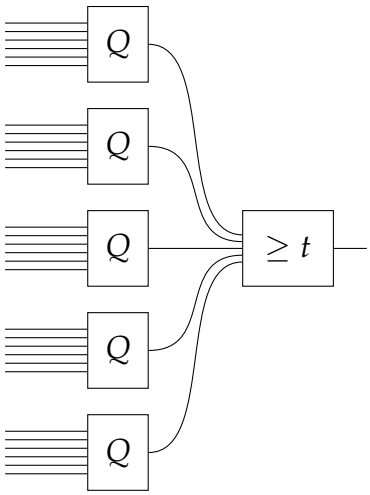


Figure 9.1: The circuit R of Construction 9.1, for $N = 5$. The box labelled $\geq t$ performs a threshold computation, outputting 1 if and only if at least t of its inputs are equal to 1.

Proof. Let $A = (A_{\text{yes}}, A_{\text{no}})$ be a promise problem contained in $\text{QMA}_p(\alpha, \beta)$, which implies that there exists a polynomial-time uniform family $\{Q_x : x \in \Sigma^*\}$ of quantum circuits satisfying the implications (8.14) and (8.15) of Definition 8.9.

Now let $\{R_x : x \in \Sigma^*\}$ be the family obtained by applying Construction 9.1 to each circuit Q_x, for $N = 2rs^2$ and

$$t = \frac{\alpha + \beta}{2} \cdot N. \tag{9.2}$$

This family is polynomial-time uniform: each R_x consists of N copies of Q_x, which can be generated in polynomial time, and the threshold t can also be computed in polynomial time by the assumption that this is true of α and β. Finally, the threshold computation itself can be performed by a circuit having size linear in N, which can also be generated in polynomial time. At this point we will fix an input string x for the remainder of the proof, writing $Q = Q_x$ and $R = R_x$.

To analyze R, let us first consider the nature of Arthur's verification circuit Q from a matrix theory viewpoint. For a given witness state ρ on p qubits sent by Merlin, Arthur accepts with probability $\langle 1|Q(\rho)|1\rangle$ and rejects with probability $\langle 0|Q(\rho)|0\rangle$. This is a binary-valued measurement of ρ, and can therefore be described by a pair of positive semidefinite matrices summing to the identity. Specifically, in the notation introduced after Definition 8.9, there must exist $2^p \times 2^p$ pos-

itive semidefinite matrices P_0 and P_1 satisfying $P_0 + P_1 = \mathbb{1}$ such that

$$\Pr(Q \text{ accepts } \rho) = \langle 1|Q(\rho)|1\rangle = \mathrm{Tr}(P_1\rho), \tag{9.3}$$

$$\Pr(Q \text{ rejects } \rho) = \langle 0|Q(\rho)|0\rangle = \mathrm{Tr}(P_0\rho), \tag{9.4}$$

for every p-qubit witness ρ. Moreover, the maximum probability with which Merlin can cause Arthur to accept on a given input string x has a simple expression:

$$\max_{\rho} \langle 1|Q(\rho)|1\rangle = \max_{\rho} \mathrm{Tr}(P_1\rho) = \lambda_{\max}(P_1), \tag{9.5}$$

the largest eigenvalue of P_1. The matrices P_0 and P_1 are uniquely determined by Q and can be expressed in different ways, perhaps most simply as

$$P_1 = Q^*(|1\rangle\langle 1|) \qquad \text{and} \qquad P_0 = Q^*(|0\rangle\langle 0|), \tag{9.6}$$

where Q^* refers to the so-called *adjoint mapping* to the channel described by Q. The adjoint of a quantum channel is a completely positive map, but it is generally not trace preserving. Rather, it is a *unital* map, which means that it maps the identity matrix (on a single qubit in this case) to the identity matrix (on p qubits in this case).

Along similar lines, the circuit R derived from Construction 9.1 with the parameters specified above gives rise to a measurement described by two positive semidefinite matrices that sum to the identity, this time on Np qubits. This measurement is equivalent to measuring each witness register independently and accepting when the number of 1s output is at least the threshold t, and therefore the matrix corresponding to acceptance may be written

$$\sum_{\substack{y \in \{0,1\}^N \\ |y|_1 \geq t}} P_{y_1} \otimes \cdots \otimes P_{y_N}. \tag{9.7}$$

(The matrix corresponding to rejection is similar, except that the sum is over all strings y with Hamming weight less than t, but we will not need to consider this matrix.) It is important to note that the matrix (9.7) determines the probability of acceptance for *every* possible Np-qubit witness, not just tensor products of N independent p-qubit witnesses.

Our goal at this point is to determine the largest eigenvalue of the matrix (9.7); along similar lines to (9.5) for the original circuit Q, this largest eigenvalue is the maximum probability that R accepts, over all Np-qubit witnesses sent by Merlin. With this goal in mind, let

$$\{|\psi_1\rangle, \ldots, |\psi_{2^p}\rangle\} \tag{9.8}$$

be an orthonormal basis of eigenvectors of P_1, with corresponding eigenvalues $\lambda_{\max}(P_1) = \lambda_1 \geq \cdots \geq \lambda_{2^p} \geq 0$. Moreover, as $P_0 = \mathbb{1}^{\otimes p} - P_1$, we have that (9.8) is also an orthonormal basis of eigenvectors of P_0 with corresponding eigenvalues $1 - \lambda_1 \leq \cdots \leq 1 - \lambda_{2^p}$. It follows that, for every choice of indices

$$(j_1, \ldots, j_N) \in \{1, \ldots, 2^p\}^N, \tag{9.9}$$

the product vector $|\psi_{j_1}\rangle \otimes \cdots \otimes |\psi_{j_N}\rangle$ is an eigenvector of the matrix (9.7) having corresponding eigenvalue $f(\lambda_{j_1}, \ldots, \lambda_{j_N})$, for

$$f(\mu_1, \ldots, \mu_N) = \sum_{\substack{y \in \{0,1\}^N \\ |y|_1 \geq t}} \prod_{k=1}^{N} \mu_k^{y_k} (1 - \mu_k)^{1 - y_k}. \tag{9.10}$$

This accounts for all 2^{pN} eigenvalues of the matrix (9.7).

The function f is non-decreasing in each of its arguments on the interval $[0, 1]$. One way to see this is to evaluate the partial derivatives by splitting the sum according to whether the corresponding bit of y is 1 or 0. For instance,

$$\frac{\partial}{\partial \mu_N} f(\mu_1, \ldots, \mu_N) = \sum_{\substack{z \in \{0,1\}^{N-1} \\ |z|_1 = \lceil t \rceil - 1}} \prod_{k=1}^{N-1} \mu_k^{z_k} (1 - \mu_k)^{1 - z_k}, \tag{9.11}$$

which is nonnegative for $\mu_1, \ldots, \mu_N \in [0, 1]$. The other partial derivatives are similar, though the indexing makes the value slightly less convenient to write. (In fact, just this one partial derivative is sufficient given that f is symmetric in its arguments.) Thus, the largest eigenvalue of the matrix (9.7) is

$$f(\lambda_1, \ldots, \lambda_1) = \sum_{\substack{y \in \{0,1\}^N \\ |y|_1 \geq t}} \lambda_1^{|y|_1} (1 - \lambda_1)^{N - |y|_1}, \tag{9.12}$$

which is the eigenvalue corresponding to the eigenvector $|\psi_1\rangle^{\otimes N}$.

The final step is to use the amplification lemma (Lemma 2.14) to establish error bounds. In particular, let $Y_1, \ldots, Y_N$ be independent random variables, each taking the value 1 with probability $\lambda_1 = \lambda_{\max}(P_1)$ and the value 0 otherwise. Each Y_k has expectation $\lambda_{\max}(P_1)$, the assumption $\alpha - \beta \geq 1/s$ gives

$$N = 2rs^2 \;\geq\; \frac{2r}{(\alpha - \beta)^2}, \tag{9.13}$$

and the value (9.12) is precisely $\Pr(Y_1 + \cdots + Y_N \geq t)$. Thus, if $x \in A_{\text{yes}}$, then $\lambda_{\max}(P_1) \geq \alpha$, so by (2.31) we have

$$\Pr(Y_1 + \cdots + Y_N \geq t) \geq 1 - 2^{-r}; \tag{9.14}$$

while if $x \in A_{\text{no}}$, then $\lambda_{\max}(P_1) \leq \beta$, so by (2.32) we have

$$\Pr(Y_1 + \cdots + Y_N \geq t) \leq 2^{-r}. \tag{9.15}$$

Therefore $A \in \text{QMA}_q(1 - 2^{-r}, 2^{-r})$, as required. □

9.2 In-place error reduction

Next we will consider in-place error reduction for QMA. As was suggested at the start of the chapter, this method makes use of unitary dilations of Arthur's verification circuits, which are run both forward and in reverse during the procedure.

Measurements from a dilated circuit

With this idea in mind, suppose that Q is a quantum circuit having p input qubits and one output qubit, representing one member of a family that describes Arthur's verification procedure, for some fixed input string x, and let U be the unitary circuit on $p + q$ qubits obtained by dilating Q. Thus, for a given pure state witness $|\psi\rangle$ on p qubits, the probability that Q accepts the witness $|\psi\rangle$ is given by

$$\begin{aligned}
\langle 1|Q(|\psi\rangle\langle\psi|)|1\rangle &= \big\| \big(\langle 1| \otimes \mathbb{1}^{\otimes(p+q-1)}\big) U\big(|\psi\rangle \otimes |0^q\rangle\big) \big\|^2 && (9.16)\\
&= \big(\langle\psi| \otimes \langle 0^q|\big) U^* \big(|1\rangle\langle 1| \otimes \mathbb{1}^{\otimes(p+q-1)}\big) U\big(|\psi\rangle \otimes |0^q\rangle\big) && (9.17)\\
&= \langle\psi|P_1|\psi\rangle && (9.18)
\end{aligned}$$

for

$$P_1 = \big(\mathbb{1}^{\otimes p} \otimes \langle 0^q|\big) U^* \big(|1\rangle\langle 1| \otimes \mathbb{1}^{\otimes(p+q-1)}\big) U\big(\mathbb{1}^{\otimes p} \otimes |0^q\rangle\big). \tag{9.19}$$

Likewise, the probability Q rejects $|\psi\rangle$ is

$$\langle 0|Q(|\psi\rangle\langle\psi|)|0\rangle = \langle\psi|P_0|\psi\rangle \tag{9.20}$$

for

$$P_0 = \big(\mathbb{1}^{\otimes p} \otimes \langle 0^q|\big) U^* \big(|0\rangle\langle 0| \otimes \mathbb{1}^{\otimes(p+q-1)}\big) U\big(\mathbb{1}^{\otimes p} \otimes |0^q\rangle\big). \tag{9.21}$$

These are precisely the matrices $P_1 = Q^*(|1\rangle\langle 1|)$ and $P_0 = Q^*(|0\rangle\langle 0|)$ that describe the action of Q as a binary-valued measurement that we encountered in the previous section; both are positive semidefinite and they sum to the identity.

In addition to the unitary circuit U, two types of measurements will be required during the procedure, both of which must be performed nondestructively, meaning that we consider not only the measurement outcomes but also make use of the post-measurement states. These measurements will be described as unitary operations within the procedure, with the measurement results being XORed onto initialized qubits, which are processed on the final step. The first is a standard basis measurement of the output qubit of Q, which we are assuming is the first (or top) qubit of U, and the second measurement is a measurement of the lower q ancillary qubits, described by the two projections $|0^q\rangle\langle 0^q|$ and $\mathbb{1}^{\otimes q} - |0^q\rangle\langle 0^q|$. In words, this measurement tests whether the ancillary qubits of U are in their initial all-zero state or not. Critically, states for which the ancillary qubits are not in the all-zero state are not disturbed by this measurement.

The in-place repetition procedure

The procedure itself is described in the following construction.

Construction 9.3. Let U be a unitary quantum circuit acting on $p + q$ qubits, let N be a positive integer, and let t be a real number. Define a quantum circuit R that inputs a p-qubit witness register W and outputs one qubit as follows.

1. Initialize a q-qubit register Y to the all-zero state.
2. For each $j = 1, \ldots, N$:
   a) Apply U to the pair of registers (W, Y).
   b) Measure the first qubit in the standard basis, letting $a_j \in \{0, 1\}$ denote the outcome.
   c) Apply U^* to the $p + q$ qubits output by U, obtaining registers (W, Y).
   d) Measure Y with the measurement
   $$\big\{|0^q\rangle\langle 0^q|,\ \mathbb{1}^{\otimes q} - |0^q\rangle\langle 0^q|\big\}, \tag{9.22}$$
   letting $b_j \in \{0, 1\}$ denote the outcome, where $b_j = 0$ corresponds to the projection $|0^q\rangle\langle 0^q|$ and $b_j = 1$ corresponds to $\mathbb{1}^{\otimes q} - |0^q\rangle\langle 0^q|$.
3. Let $y = y_1 \cdots y_{2N} \in \{0, 1\}^{2N}$ be the string defined by
   $$y_1 = a_1, \qquad y_{2j} = a_j \oplus b_j, \qquad y_{2j-1} = a_j \oplus b_{j-1}, \tag{9.23}$$
   for $j = 1, \ldots, N$ in the second case and $j = 2, \ldots, N$ in the third.
4. Accept if $|y|_1 \geq t$, and reject otherwise.

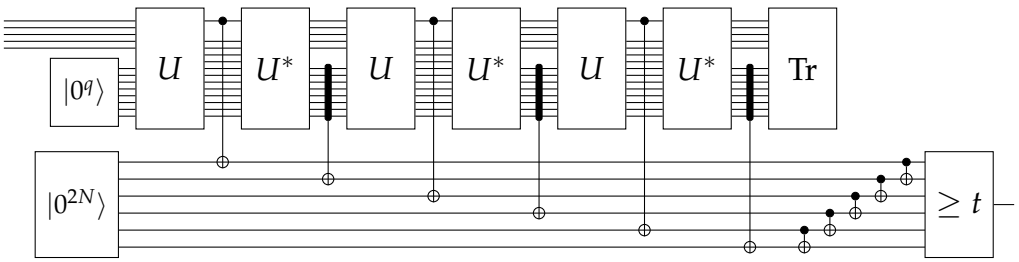


Figure 9.2: The circuit R described by Construction 9.3 when $N = 3$. The tall ovals spanning the ancillary qubits are to be interpreted as OR-controls, meaning that the target qubit is flipped if and only if at least one of the q control qubits is 1. The controlled-NOT gates on the right combine the recorded outcomes into the string y.

Main theorem

An analysis of Construction 9.3 reveals the following theorem.

Theorem 9.4 (Marriott–Watrous). *Let $\alpha, \beta : \mathbb{N} \to [0,1]$ be polynomial-time computable functions such that $\alpha(n) - \beta(n) \geq 1/s(n)$ for every n for a polynomial resource bound s, and let p and r be given polynomial resource bounds. Then*

$$\mathrm{QMA}_p(\alpha, \beta) \subseteq \mathrm{QMA}_p\bigl(1 - 2^{-r}, 2^{-r}\bigr). \tag{9.24}$$

Proof. Let $A = (A_{\mathrm{yes}}, A_{\mathrm{no}})$ be a promise problem contained in $\mathrm{QMA}_p(\alpha, \beta)$, which implies that there exists a polynomial-time uniform family $\{Q_x : x \in \Sigma^*\}$ of quantum circuits satisfying the implications (8.14) and (8.15) of Definition 8.9. Also let $\{U_x : x \in \Sigma^*\}$ be a polynomial-time uniform family of unitary quantum circuits, as provided by Proposition 5.2, such that each U_x is a dilation of Q_x that uses $q = q(|x|)$ ancillary qubits, for q a polynomial resource bound. Finally, let $\{R_x : x \in \Sigma^*\}$ be the family of quantum circuits obtained by applying Construction 9.3 to each U_x, taking

$$N = rs^2 \qquad \text{and} \qquad t = (\alpha + \beta)N. \tag{9.25}$$

This family is polynomial-time uniform, as each R_x is assembled from N copies each of U_x and U_x^* together with a circuit performing the threshold value computation. For the remainder of the proof we will assume that an input string x has been fixed, and we will write $U = U_x$ and $R = R_x$.

The principal goal for the remainder of the proof is to analyze the maximum probability with which R accepts over all choices of a witness state. We will first

consider the performance of R when the witness is a pure state $|\psi\rangle$ that is an eigenvector of P_1 with eigenvalue λ. This implies that Q accepts $|\psi\rangle$ with probability λ. Also, because $P_0 = \mathbb{1}^{\otimes p} - P_1$, we have that $|\psi\rangle$ is an eigenvector of P_0 with eigenvalue $1 - \lambda$.

Define a unit vector

$$|\phi_0\rangle = |\psi\rangle \otimes |0^q\rangle, \tag{9.26}$$

which represents the initial state of the top $p + q$ qubits of R, along with projections on $p + q$ qubits as follows.

$$\Pi_0 = U^* \big(|0\rangle\langle 0| \otimes \mathbb{1}^{\otimes (p+q-1)} \big) U \tag{9.27}$$

$$\Pi_1 = U^* \big(|1\rangle\langle 1| \otimes \mathbb{1}^{\otimes (p+q-1)} \big) U \tag{9.28}$$

$$\Delta_0 = \mathbb{1}^{\otimes p} \otimes |0^q\rangle\langle 0^q| \tag{9.29}$$

$$\Delta_1 = \mathbb{1}^{\otimes p} \otimes \big(\mathbb{1}^{\otimes q} - |0^q\rangle\langle 0^q| \big) \tag{9.30}$$

In effect, starting from the state $|\phi_0\rangle$, the circuit R measures the $p + q$ qubits alternately with respect to $\{\Pi_0, \Pi_1\}$ and with respect to $\{\Delta_0, \Delta_1\}$, with $a_1, b_1, \ldots, a_N, b_N$ denoting the results.

We will determine the probabilities of each sequence and the successive states of the $p + q$ qubits conditioned on each outcome. To do this, we begin with a few observations. First, it is clear that $\Delta_0|\phi_0\rangle = |\phi_0\rangle$ and $\Delta_1|\phi_0\rangle = 0$. Moreover,

$$\Delta_0 \Pi_0 \Delta_0 = P_0 \otimes |0^q\rangle\langle 0^q|, \tag{9.31}$$

$$\Delta_0 \Pi_1 \Delta_0 = P_1 \otimes |0^q\rangle\langle 0^q|, \tag{9.32}$$

from which it follows that

$$\Delta_0 \Pi_0 |\phi_0\rangle = \Delta_0 \Pi_0 \Delta_0 |\phi_0\rangle = (1 - \lambda)|\phi_0\rangle, \tag{9.33}$$

$$\Delta_0 \Pi_1 |\phi_0\rangle = \Delta_0 \Pi_1 \Delta_0 |\phi_0\rangle = \lambda |\phi_0\rangle. \tag{9.34}$$

The first measurement is $\{\Pi_0, \Pi_1\}$, performed on $|\phi_0\rangle$. We have

$$\Pr(a_1 = 0) = \|\Pi_0|\phi_0\rangle\|^2 = \|\Pi_0 \Delta_0 |\phi_0\rangle\|^2 = \langle \phi_0 | \Delta_0 \Pi_0 \Delta_0 | \phi_0 \rangle = 1 - \lambda, \tag{9.35}$$

$$\Pr(a_1 = 1) = \|\Pi_1|\phi_0\rangle\|^2 = \|\Pi_1 \Delta_0 |\phi_0\rangle\|^2 = \langle \phi_0 | \Delta_0 \Pi_1 \Delta_0 | \phi_0 \rangle = \lambda, \tag{9.36}$$

and the states of the $p+q$ qubits conditioned on these outcomes, which we will name $|\chi_0\rangle$ and $|\chi_1\rangle$, are the normalizations of these vectors:[1]

$$|\chi_0\rangle = \frac{\Pi_0|\phi_0\rangle}{\sqrt{1-\lambda}} \qquad \text{and} \qquad |\chi_1\rangle = \frac{\Pi_1|\phi_0\rangle}{\sqrt{\lambda}}. \tag{9.37}$$

Next we will analyze what happens when $|\chi_0\rangle$ and $|\chi_1\rangle$ are measured with respect to $\{\Delta_0, \Delta_1\}$. The easier projection to handle is Δ_0, for which we have

$$\Delta_0|\chi_0\rangle = \frac{\Delta_0\Pi_0|\phi_0\rangle}{\sqrt{1-\lambda}} = \frac{\Delta_0\Pi_0\Delta_0|\phi_0\rangle}{\sqrt{1-\lambda}} = \sqrt{1-\lambda}|\phi_0\rangle, \tag{9.38}$$

$$\Delta_0|\chi_1\rangle = \frac{\Delta_0\Pi_1|\phi_0\rangle}{\sqrt{\lambda}} = \frac{\Delta_0\Pi_1\Delta_0|\phi_0\rangle}{\sqrt{\lambda}} = \sqrt{\lambda}|\phi_0\rangle. \tag{9.39}$$

To analyze the action of the projection Δ_1 on $|\chi_0\rangle$ and $|\chi_1\rangle$, notice first that

$$\Delta_1\big(\sqrt{1-\lambda}|\chi_0\rangle + \sqrt{\lambda}|\chi_1\rangle\big) = \Delta_1|\phi_0\rangle = 0, \tag{9.40}$$

so

$$\sqrt{1-\lambda}\Delta_1|\chi_0\rangle = -\sqrt{\lambda}\Delta_1|\chi_1\rangle. \tag{9.41}$$

With this in mind, let us define a new unit vector

$$|\phi_1\rangle = \frac{\Delta_1|\chi_0\rangle}{\sqrt{\lambda}} = -\frac{\Delta_1|\chi_1\rangle}{\sqrt{1-\lambda}}, \tag{9.42}$$

so that

$$\Delta_1|\chi_0\rangle = \sqrt{\lambda}|\phi_1\rangle, \tag{9.43}$$

$$\Delta_1|\chi_1\rangle = -\sqrt{1-\lambda}|\phi_1\rangle. \tag{9.44}$$

The fact that $|\phi_1\rangle$ is a unit vector follows from $\|\Delta_0|\chi_0\rangle\|^2 + \|\Delta_1|\chi_0\rangle\|^2 = 1$, implying

$$\|\Delta_1|\chi_0\rangle\|^2 = 1 - \|\Delta_0|\chi_0\rangle\|^2 = 1 - (1-\lambda) = \lambda. \tag{9.45}$$

This new vector $|\phi_1\rangle$ represents the state of the $p+q$ qubits conditioned on the measurement $\{\Delta_0, \Delta_1\}$ outputting 1, so it is entirely supported on standard basis states for which the bottom q ancillary qubits are not all zero.

[1] Note that when λ is 0 or 1, one of the two denominators is zero, so the corresponding vector is undefined. This causes no difficulty and the undefined vector can safely be disregarded, as we need not be concerned with states conditioned on outcomes having probability zero. In more concrete terms, if $\lambda = 0$, then every measurement produces the outcome 0, and if $\lambda = 1$, then every measurement produces $a_j = 1$ and $b_j = 0$. In both cases, the state of the $p+q$ qubits is $|\phi_0\rangle$ at the end of the procedure.

In summary, measuring $|\chi_0\rangle$ with respect to $\{\Delta_0, \Delta_1\}$ gives outcomes 0 and 1 with probabilities $1-\lambda$ and λ, and conditioned on these outcomes the resulting state becomes $|\phi_0\rangle$ or $|\phi_1\rangle$, respectively; while a measurement of $|\chi_1\rangle$ with respect to $\{\Delta_0, \Delta_1\}$ gives outcomes 0 and 1 with probabilities λ and $1-\lambda$ (which are *swapped* in comparison to measuring $|\chi_0\rangle$), and conditioned on these outcomes the resulting states become $|\phi_0\rangle$ or $|\phi_1\rangle$, respectively (which are the *same* as when measuring $|\chi_0\rangle$). That is,

$$\Pr(b_1 = 0 \mid a_1 = 0) = 1 - \lambda, \tag{9.46}$$

$$\Pr(b_1 = 1 \mid a_1 = 0) = \lambda, \tag{9.47}$$

$$\Pr(b_1 = 0 \mid a_1 = 1) = \lambda, \tag{9.48}$$

$$\Pr(b_1 = 1 \mid a_1 = 1) = 1 - \lambda, \tag{9.49}$$

with the resulting state of the $p+q$ qubits always being $|\phi_0\rangle$ when $b_1 = 0$ and $|\phi_1\rangle$ when $b_1 = 1$.

Subsequent iterations follow the same pattern when starting from the state $|\phi_0\rangle$, so it remains to consider what happens starting from $|\phi_1\rangle$. Fortunately, we already have enough information from the above analysis to determine what happens in this case. In particular, using (9.42), along with $\Delta_0 + \Delta_1 = \mathbb{1}^{\otimes(p+q)}$, we have

$$\Pi_0|\phi_1\rangle = \frac{\Pi_0\Delta_1|\chi_0\rangle}{\sqrt{\lambda}} = \frac{|\chi_0\rangle - \Pi_0\Delta_0|\chi_0\rangle}{\sqrt{\lambda}} = \sqrt{\lambda}|\chi_0\rangle, \tag{9.50}$$

$$\Pi_1|\phi_1\rangle = -\frac{\Pi_1\Delta_1|\chi_1\rangle}{\sqrt{1-\lambda}} = -\frac{|\chi_1\rangle - \Pi_1\Delta_0|\chi_1\rangle}{\sqrt{1-\lambda}} = -\sqrt{1-\lambda}|\chi_1\rangle. \tag{9.51}$$

Thus, measuring $|\phi_1\rangle$ with respect to $\{\Pi_0, \Pi_1\}$ gives outcomes 0 and 1 with probabilities λ and $1-\lambda$, and conditioned on these outcomes the resulting state becomes $|\chi_0\rangle$ or $|\chi_1\rangle$, respectively.

The pattern repeats for successive iterations. The state of the $p+q$ qubits immediately before a measurement with respect to $\{\Pi_0, \Pi_1\}$ is always $|\phi_0\rangle$ or $|\phi_1\rangle$, and immediately before a measurement with respect to $\{\Delta_0, \Delta_1\}$ it is always $|\chi_0\rangle$ or $|\chi_1\rangle$. The outcome probabilities and resulting states have therefore already been determined above. We obtain

$$\Pr(a_{j+1} = 0 \mid b_j = 0) = 1 - \lambda, \tag{9.52}$$

$$\Pr(a_{j+1} = 1 \mid b_j = 0) = \lambda, \tag{9.53}$$

$$\Pr(a_{j+1} = 0 \mid b_j = 1) = \lambda, \tag{9.54}$$

$$\Pr(a_{j+1} = 1 \mid b_j = 1) = 1 - \lambda, \tag{9.55}$$

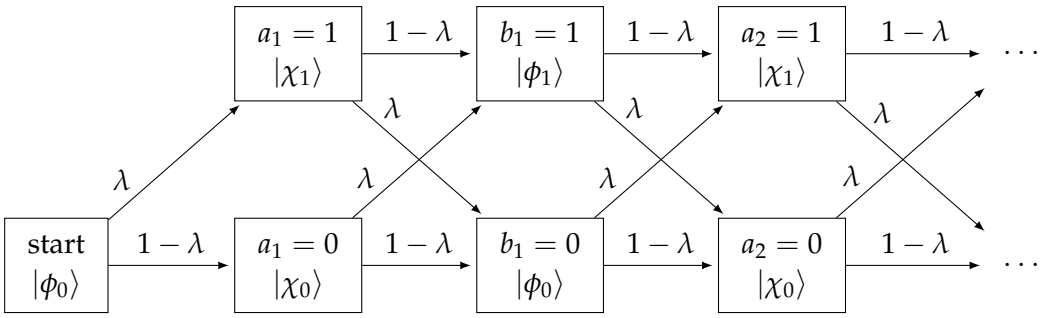


Figure 9.3: Transition probabilities between consecutive measurement outcomes, along with the states conditioned on these outcomes, in the in-place error reduction procedure, assuming an initial witness state that is an eigenvector of P_1 having eigenvalue λ. The measurement outcomes flip with probability λ and remain the same with probability $1 - \lambda$.

with the resulting state of the $p + q$ qubits always being $|\chi_0\rangle$ when $a_{j+1} = 0$ and $|\chi_1\rangle$ when $a_{j+1} = 1$. In the same way, the conditional probabilities (9.46)–(9.49) generalize to

$$\Pr(b_j = 0 \mid a_j = 0) = 1 - \lambda, \tag{9.56}$$

$$\Pr(b_j = 1 \mid a_j = 0) = \lambda, \tag{9.57}$$

$$\Pr(b_j = 0 \mid a_j = 1) = \lambda, \tag{9.58}$$

$$\Pr(b_j = 1 \mid a_j = 1) = 1 - \lambda, \tag{9.59}$$

for all $j = 1, \ldots, N$, and once again the resulting state of the $p + q$ qubits becomes $|\phi_0\rangle$ when $b_j = 0$ and $|\phi_1\rangle$ when $b_j = 1$. Figure 9.3 expresses the pattern in the form of a diagram.

The critical take-away is that each measurement result agrees with the previous one with probability $1 - \lambda$ and disagrees with probability λ, with the first measurement result being 1 with probability λ and 0 with probability $1 - \lambda$. As $y_1 = a_1$, and $y_2, \ldots, y_{2N}$ indicate whether or not successive measurement results disagree, we find that $y_1, \ldots, y_{2N}$ are described by independent binary-valued random variables, each having expected value λ. Thus, R accepts with probability $f(\lambda)$ for

$$f(\mu) = \sum_{\substack{y \in \{0,1\}^{2N} \\ |y|_1 \geq t}} \mu^{|y|_1} (1 - \mu)^{2N - |y|_1}, \tag{9.60}$$

which is the probability that $2N$ independent binary-valued random variables, each equal to 1 with probability μ, sum to at least t.

The last thing we need to do is to consider the case of a general witness, not necessarily an eigenvector of P_0 and P_1. It suffices to focus on pure states, which reduces to the eigenvector case when a witness is expressed as a linear combination of eigenvectors. With this in mind, let

$$\{|\psi_1\rangle, \ldots, |\psi_{2^p}\rangle\} \tag{9.61}$$

be an orthonormal basis of unit eigenvectors of P_1 having corresponding eigenvalues $\lambda_1, \ldots, \lambda_{2^p}$, and for each $k \in \{1, \ldots, 2^p\}$ let us write

$$|\phi_0^k\rangle, \quad |\phi_1^k\rangle, \quad |\chi_0^k\rangle, \quad |\chi_1^k\rangle \tag{9.62}$$

to denote the corresponding states in the analysis of the eigenvector case above, taking $|\psi\rangle = |\psi_k\rangle$ and $\lambda = \lambda_k$. (As before, some of these vectors are undefined when λ_k is 0 or 1, but once again they will not be needed.)

The key step required at this point is to establish that

$$\{|\phi_a^k\rangle \,:\, a \in \{0,1\},\ k \in \{1, \ldots, 2^p\}\} \tag{9.63}$$

is an orthonormal set. First, it is clear that $\langle \phi_0^i | \phi_0^k \rangle = 0$ whenever $i \neq k$ by the orthogonality of (9.61). Next, by combining

$$|\phi_1^k\rangle = \frac{\Delta_1 |\chi_0^k\rangle}{\sqrt{\lambda_k}} = \frac{|\chi_0^k\rangle - \Delta_0 |\chi_0^k\rangle}{\sqrt{\lambda_k}} \tag{9.64}$$

with

$$|\chi_0^k\rangle = \frac{\Pi_0 |\phi_0^k\rangle}{\sqrt{1-\lambda_k}} = \frac{\Pi_0 \Delta_0 |\phi_0^k\rangle}{\sqrt{1-\lambda_k}}, \tag{9.65}$$

we obtain (after just a bit of simplification)

$$\langle \phi_1^i | \phi_1^k \rangle = \frac{\langle \phi_0^i | \Delta_0 \Pi_0 \Delta_0 | \phi_0^k \rangle - \langle \phi_0^i | (\Delta_0 \Pi_0 \Delta_0)^2 | \phi_0^k \rangle}{\sqrt{\lambda_i (1-\lambda_i) \lambda_k (1-\lambda_k)}} = 0 \tag{9.66}$$

for every $i \neq k$, owing to $\Delta_0 \Pi_0 \Delta_0 = P_0 \otimes |0^q\rangle\langle 0^q|$ and $|\phi_0^k\rangle = |\psi_k\rangle \otimes |0^q\rangle$. Finally, $\langle \phi_0^i | \phi_1^k \rangle = 0$ for all i and k because $\Delta_0 \Delta_1 = 0$.

Lastly, consider a unitary dilation of the circuit in Figure 9.2 with the threshold computation removed, which we will call V — so V is a unitary circuit acting on $p + q + 2N$ qubits. We have already established that running V on the state $|\phi_0^k\rangle \otimes |0^{2N}\rangle$ results in a state of the form

$$V(|\phi_0^k\rangle \otimes |0^{2N}\rangle) = |\phi_0^k\rangle \otimes |\eta_0^k\rangle + |\phi_1^k\rangle \otimes |\eta_1^k\rangle \tag{9.67}$$

for $2N$-qubit vectors $|\eta_0^k\rangle$ and $|\eta_1^k\rangle$ that satisfy

$$|\langle y|\eta_0^k\rangle|^2 + |\langle y|\eta_1^k\rangle|^2 = \lambda_k^{|y|_1}(1-\lambda_k)^{2N-|y|_1} \tag{9.68}$$

for every $y \in \{0,1\}^{2N}$, which is the probability that a standard basis measurement of the lower $2N$ qubits yields y (which goes into the threshold computation in R). By linearity, for a general pure state witness

$$|\psi\rangle = \sum_{k=1}^{2^p} \gamma_k |\psi_k\rangle \tag{9.69}$$

we therefore have

$$V\big(|\psi\rangle \otimes |0^q\rangle \otimes |0^{2N}\rangle\big) = \sum_{k=1}^{2^p} \gamma_k \big(|\phi_0^k\rangle \otimes |\eta_0^k\rangle + |\phi_1^k\rangle \otimes |\eta_1^k\rangle\big). \tag{9.70}$$

The top $p+q$ qubits are traced out in R, which leaves the lower $2N$ qubits in the state

$$\sum_{k=1}^{2^p} |\gamma_k|^2 \Big(|\eta_0^k\rangle\langle\eta_0^k| + |\eta_1^k\rangle\langle\eta_1^k|\Big). \tag{9.71}$$

This is a convex combination of the states that these qubits would have been in had Merlin sent each eigenvector $|\psi_k\rangle$ with probability $|\gamma_k|^2$. The circuit R therefore accepts $|\psi\rangle$ with probability

$$\sum_k |\gamma_k|^2 f(\lambda_k). \tag{9.72}$$

The final step in the proof is an application of Lemma 2.14 (the amplification lemma). The assumption $\alpha - \beta \geq 1/s$ implies that

$$2N = 2rs^2 \;\geq\; \frac{2r}{(\alpha-\beta)^2}, \tag{9.73}$$

while

$$t = (\alpha+\beta)N = \frac{\alpha+\beta}{2} \cdot 2N, \tag{9.74}$$

is the midpoint threshold between $2\alpha N$ and $2\beta N$. If $x \in A_{\text{yes}}$, then P_1 has an eigenvector whose eigenvalue is $\lambda_{\max}(P_1) \geq \alpha$, so (2.31) shows that R accepts this witness with probability at least $1 - 2^{-r}$. If $x \in A_{\text{no}}$, then every eigenvalue of P_1 is at most $\lambda_{\max}(P_1) \leq \beta$, so (2.32) gives $f(\lambda_k) \leq 2^{-r}$ for every k, so the probability (9.72) with which any given witness state is accepted is at most 2^{-r}. Because the witness length p has not changed, we conclude that $A \in \text{QMA}_p(1-2^{-r}, 2^{-r})$, as required. □

9.3 Two consequences

In this section we discuss the two consequences of Theorem 9.4 mentioned earlier: the inclusion QMA $\subseteq$ PP and the observation that logarithmic-length quantum witnesses offer no advantage over BQP.

QMA is contained in PP

We will make use of the following fact: if P is an $d \times d$ positive semidefinite matrix, then

$$\lambda_{\max}(P) \leq \mathrm{Tr}(P) \leq d\lambda_{\max}(P). \tag{9.75}$$

This is because (1) the trace of every square matrix is equal to the sum of its eigenvalues, and (2) every eigenvalue of a positive semidefinite matrix is nonnegative. Hence, if

$$\lambda_{\max}(P) = \lambda_1 \geq \cdots \geq \lambda_d \geq 0 \tag{9.76}$$

are the eigenvalues of P, then clearly $\lambda_{\max}(P) \leq \mathrm{Tr}(P)$ and

$$\mathrm{Tr}(P) = \lambda_1 + \cdots + \lambda_d \leq d\lambda_1 = d\lambda_{\max}(P). \tag{9.77}$$

With this in mind, we now prove Theorem 8.12, which states that QMA $\subseteq$ PP.

Proof of Theorem 8.12. Let $A = (A_{\mathrm{yes}}, A_{\mathrm{no}}) \in \mathrm{QMA}$, so that $A \in \mathrm{QMA}_p(2/3, 1/3)$ for some polynomial resource bound p. Applying Theorem 9.4 with the polynomial resource bound $r = p + 2$, we obtain

$$A \in \mathrm{QMA}_p\big(1 - 2^{-(p+2)},\, 2^{-(p+2)}\big). \tag{9.78}$$

Take $\{Q_x\}$ to be a family as in Definition 8.9 for these parameters, and let P_x be the acceptance matrix of Q_x, as in the proof of Theorem 9.2, so that

$$\Pr(Q_x \text{ accepts } \rho) = \mathrm{Tr}(P_x\rho) \tag{9.79}$$

for every p-qubit state ρ. Using the fact mentioned above, we have these implications:

$$x \in A_{\mathrm{yes}} \implies \mathrm{Tr}(P_x) \geq \lambda_{\max}(P_x) \geq 1 - 2^{-(p+2)} \geq 3/4, \tag{9.80}$$

$$x \in A_{\mathrm{no}} \implies \mathrm{Tr}(P_x) \leq 2^p\lambda_{\max}(P_x) \leq 2^p 2^{-(p+2)} = 1/4. \tag{9.81}$$

Next, observe that

$$\mathrm{Tr}(P_x) = \sum_{w\in\{0,1\}^p} \langle w|P_x|w\rangle = \sum_{w\in\{0,1\}^p} \langle 1|Q_x\big(|w\rangle\langle w|\big)|1\rangle, \tag{9.82}$$

where the second equality follows by taking $\rho = |w\rangle\langle w|$ in (9.79). By Theorem 6.14, there exist GapP functions f and g and a polynomial resource bound s such that

$$f(x,u,v,y,z) + ig(x,u,v,y,z) = 2^s \langle y| Q_x\big(|u\rangle\langle v|\big) |z\rangle \tag{9.83}$$

for all $u, v \in \{0,1\}^p$ and $y, z \in \{0,1\}$. The diagonal entries of a density matrix are real, so the function

$$h(x) = \sum_{w \in \{0,1\}^p} f(x,w,w,1,1) \tag{9.84}$$

satisfies $h(x) = 2^s \operatorname{Tr}(P_x)$. It follows from Theorem 6.9 that $h \in \mathrm{GapP}$.

Finally, $x \mapsto 2^s$ is a #P function, so $2h(x) - 2^s$ is a GapP function. By (9.80) it is at least $\frac{1}{2} \cdot 2^s > 0$ for $x \in A_{\mathrm{yes}}$, and by (9.81) it is at most $-\frac{1}{2} \cdot 2^s < 0$ for $x \in A_{\mathrm{no}}$, so Proposition 6.8 gives $A \in \mathrm{PP}$. □

Logarithmic-length witnesses

Finally, the following theorem implies that logarithmic-length quantum witnesses offer no additional power beyond BQP.

Theorem 9.5. *Let $\alpha, \beta : \mathbb{N} \to [0,1]$ be polynomial-time computable functions for which there exists a polynomial resource bound s satisfying $\alpha(n) - \beta(n) \geq 1/s(n)$ for every n, and let p be a polynomial resource bound satisfying $p(n) = O(\log n)$. Then $\mathrm{QMA}_p(\alpha,\beta) \subseteq \mathrm{BQP}$.*

Proof. Let $A \in \mathrm{QMA}_p(\alpha,\beta)$. Applying Theorem 9.4 with the polynomial resource bound $r = p + 2$ gives $A \in \mathrm{QMA}_p(1 - 2^{-(p+2)}, 2^{-(p+2)})$. Exactly as in (9.80) and (9.81), taking $\{Q_x\}$ to be a family for these parameters and P_x its acceptance matrices, we find that $\operatorname{Tr}(P_x) \geq 3/4$ for $x \in A_{\mathrm{yes}}$ and $\operatorname{Tr}(P_x) \leq 1/4$ for $x \in A_{\mathrm{no}}$.

Now define a family $\{V_x\}$ of quantum circuits, each of which runs Q_x on the completely mixed state $\mathbb{1}^{\otimes p}/2^p$ in place of a witness, prepared using p copies of the random-bit generator of Figure 5.1. The family $\{V_x\}$ is polynomial-time uniform, and

$$\Pr(V_x \text{ accepts}) = \operatorname{Tr}\Big(P_x \frac{\mathbb{1}^{\otimes p}}{2^p}\Big) = \frac{\operatorname{Tr}(P_x)}{2^p}, \tag{9.85}$$

so V_x accepts with probability at least $\frac{3}{4} \cdot 2^{-p}$ when $x \in A_{\mathrm{yes}}$ and at most $\frac{1}{4} \cdot 2^{-p}$ when $x \in A_{\mathrm{no}}$.

As $p(n) = O(\log n)$, the function $q = 2^{p+1}$ is polynomially bounded, and is therefore a polynomial resource bound. The two bounds just obtained satisfy the requirements of Theorem 5.5 for this choice of q:

$$2^{-q} \leq \tfrac{1}{4} \cdot 2^{-p} \leq \tfrac{3}{4} \cdot 2^{-p} \leq 1 - 2^{-q} \qquad \text{and} \qquad \tfrac{3}{4} \cdot 2^{-p} - \tfrac{1}{4} \cdot 2^{-p} = \frac{1}{q}, \tag{9.86}$$

where the first inequality holds because $2^{p+1} \geq p+2$ and the third because $\frac{3}{4} \cdot 2^{-p} \leq \frac{3}{4} \leq 1 - 2^{-q}$. Both bounds are polynomial-time computable, so the theorem yields

$$A \in \mathrm{BQP}\left(\tfrac{3}{4} \cdot 2^{-p}, \tfrac{1}{4} \cdot 2^{-p}\right) = \mathrm{BQP}, \tag{9.87}$$

as required. □

Chapter 10

The local Hamiltonian problem

In this chapter we will define a problem known as the *local Hamiltonian problem* and prove that it is complete for QMA with respect to Karp reductions. This fact may, in some sense, be seen as a quantum analogue of the Cook–Levin theorem (Theorem 3.14).

10.1 Pauli matrix preliminaries

Before discussing the local Hamiltonian problem itself, we will require some basic facts concerning Pauli matrices. Recall that the *Pauli matrices* (including the identity matrix) are as follows:

$$\mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{10.1}$$

Each of these matrices is both unitary and Hermitian.

Pauli operations

For each positive integer n, define the set

$$\mathcal{P}_n = \big\{ P_1 \otimes \cdots \otimes P_n \,:\, P_1, \ldots, P_n \in \{\mathbb{1}, X, Y, Z\} \big\}, \tag{10.2}$$

which consists of 4^n matrices, each of which is $2^n \times 2^n$. We will refer to elements of the set $\mathcal{P}_n$ as n-qubit *Pauli operations*. As each of the Pauli matrices is both unitary

and Hermitian, the same is true of every n-qubit Pauli operation. The formula

$$\mathrm{Tr}(PQ) = \begin{cases} 2^n & \text{if } P = Q \\ 0 & \text{if } P \neq Q \end{cases} \tag{10.3}$$

holds for all $P, Q \in \mathcal{P}_n$.

The *weight* of an n-qubit Pauli operation is the number of indices $i \in \{1, \ldots, n\}$ for which $P_i \neq \mathbb{1}$. Intuitively speaking, a weight k Pauli operation acts trivially (as the identity) on all but k qubits.

Finally, we note that every $2^n \times 2^n$ matrix M can be expressed as a linear combination of n-qubit Pauli operations. More specifically, it is the case that

$$M = \sum_{P \in \mathcal{P}_n} \gamma_P P \qquad \text{for} \qquad \gamma_P = \frac{\mathrm{Tr}(PM)}{2^n}. \tag{10.4}$$

If M is Hermitian, then each γ_P is a real number satisfying $|\gamma_P| \leq \|M\|$. Also, the entries of a Hermitian matrix M are Gaussian rational numbers if and only if each γ_P is rational.

Measuring Pauli operations

Pauli operations can be viewed not only as unitary operations but also as measurements. In particular, with every n-qubit Pauli operation $P \in \mathcal{P}_n$ we associate a projective measurement whose possible outcomes are $+1$ and -1. The projections corresponding to these two possible outcomes are the projections onto the $+1$ and -1 eigenspaces of P, respectively.

In greater detail, because Pauli operations are both Hermitian and unitary, the only possible eigenvalues they can have are $+1$ and -1. It follows that we can always write

$$P = \Pi_+ - \Pi_- \tag{10.5}$$

for two orthogonal projections Π_+ and Π_- satisfying $\Pi_+ + \Pi_- = \mathbb{1}^{\otimes n}$. We can express these projections explicitly as

$$\Pi_+ = \frac{\mathbb{1}^{\otimes n} + P}{2} \qquad \text{and} \qquad \Pi_- = \frac{\mathbb{1}^{\otimes n} - P}{2}. \tag{10.6}$$

The measurement associated with P is the two-outcome measurement defined by these two projections, where Π_+ corresponds to the outcome $+1$ and Π_- corresponds to the outcome -1. (It is also common to view the outcomes as being 0

and 1 rather than $+1$ and -1, with the correspondence $0 \leftrightarrow +1$, $1 \leftrightarrow -1$ being implicit.) If this measurement is performed on a state ρ of n qubits, the outcomes $+1$ and -1 are obtained with probabilities

$$\mathrm{Tr}(\Pi_{\pm}\rho) = \frac{1 \pm \mathrm{Tr}(P\rho)}{2}. \tag{10.7}$$

There are two natural ways to perform such a measurement with a quantum circuit:

1. **Qubit by qubit.** One way to perform the measurement associated with an n-qubit Pauli operation $P \in \mathcal{P}_n$ is to individually perform the Pauli measurement defined by each tensor factor of P on its corresponding qubit, followed by a multiplication of the results.

   Explicitly, the single-qubit measurement corresponding to X is equivalent to measuring with respect to the $\{|+\rangle, |-\rangle\}$ basis; the measurement corresponding to Y is equivalent to measuring with respect to the $\{|+i\rangle, |-i\rangle\}$ basis, where $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$; and the measurement corresponding to Z is equivalent to measuring in the standard basis $\{|0\rangle, |1\rangle\}$, with each outcome being mapped to ± 1 accordingly. This is consistent with the spectral decompositions

   $$X = |+\rangle\langle +| - |-\rangle\langle -|, \tag{10.8}$$

   $$Y = |+i\rangle\langle +i| - |-i\rangle\langle -i|, \tag{10.9}$$

   $$Z = |0\rangle\langle 0| - |1\rangle\langle 1|. \tag{10.10}$$

   These measurements can be performed by first applying one of the unitary operations H, HS^*, or $\mathbb{1}$ (respectively) followed by a standard basis measurement (associating the outcome 0 with $+1$ and the outcome 1 with -1).

2. **Through phase estimation.** Another way to perform the measurement associated with $P \in \mathcal{P}_n$ is to use phase estimation, which requires a single control qubit because the eigenvalues of P are all either $+1$ or -1. This is equivalent to using the controlled-unitary test shown in Figure 8.2 for $U = P$. Once again, the measurement outputs 0 and 1 are associated with the outcomes $+1$ and -1, respectively. For example, the circuit in Figure 10.1 performs the measurement associated with $P = X \otimes Z \otimes Y$.

In some settings (such as quantum error correction), it is important to use the second method rather than the first so that $+1$ and -1 eigenstates of P are not

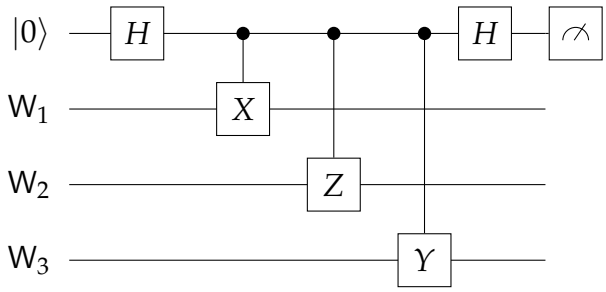


Figure 10.1: A circuit performing the measurement associated with $X \otimes Z \otimes Y$ on three qubits $(\mathsf{W}_1, \mathsf{W}_2, \mathsf{W}_3)$. It is the controlled-unitary test of Figure 8.2 for $U = X \otimes Z \otimes Y$, with the controlled-$U$ operation performed as three controlled Pauli gates sharing a control qubit. The outcomes 0 and 1 correspond to $+1$ and -1, respectively.

disturbed by the measurement. This is not critical for this chapter, but the phase estimation approach does have a different applicable advantage: it is much simpler to perform the measurements exactly with the standard gate set using this method, even when the measurements themselves are controlled (e.g., when other qubits in a circuit determine which of a collection of Pauli measurements to perform).

In particular, writing $\Lambda(U)$ to denote a controlled-U operation in general, the controlled X, Y, and Z gates needed for phase estimation can be realized as

$$\begin{aligned} \Lambda(X) &= \text{CNOT}, \\ \Lambda(Y) &= (\mathbb{1} \otimes S)\,\text{CNOT}\,(\mathbb{1} \otimes S^*), \\ \Lambda(Z) &= (\mathbb{1} \otimes H)\,\text{CNOT}\,(\mathbb{1} \otimes H), \end{aligned} \tag{10.11}$$

while doubly controlled versions of these gates (which can be used to select one of a collection of Pauli measurements to perform) are realized as

$$\begin{aligned} \Lambda(\Lambda(X)) &= \text{TOF}, \\ \Lambda(\Lambda(Y)) &= (\mathbb{1} \otimes \mathbb{1} \otimes S)\,\text{TOF}\,(\mathbb{1} \otimes \mathbb{1} \otimes S^*), \\ \Lambda(\Lambda(Z)) &= (\mathbb{1} \otimes \mathbb{1} \otimes H)\,\text{TOF}\,(\mathbb{1} \otimes \mathbb{1} \otimes H). \end{aligned} \tag{10.12}$$

10.2 Problem definition

Now we are ready to define the local Hamiltonian problem. There are different (but equivalent) ways to define this problem; our definition will be in terms of linear combinations of Pauli operations.

Local Hamiltonians

First we define what is meant by a k-local Hamiltonian for any positive integer k.

Definition 10.1. Let n and k be positive integers with $k \leq n$. A $2^n \times 2^n$ Hermitian matrix H (i.e., a Hermitian matrix acting on an n-qubit state space) is a *k-local Hamiltonian* if

$$H = \gamma_1 P_1 + \cdots + \gamma_m P_m \tag{10.13}$$

for real numbers $\gamma_1, \ldots, \gamma_m$ and Pauli operations $P_1, \ldots, P_m \in \mathcal{P}_n$, each having weight at most k.

This is not the most typical way to define k-local Hamiltonians — but it is equivalent and (arguably) simpler to state formally. The more typical definition is that a k-local Hamiltonian is a Hermitian matrix of the form $H = H_1 + \cdots + H_m$ where each H_j acts non-trivially on at most k qubits. Technically speaking, this means that up to a permutation of the qubits, each H_j takes the form $M_j \otimes \mathbb{1}^{\otimes(n-k)}$, where each M_j is a $2^k \times 2^k$ Hermitian matrix (i.e., M_j acts on k qubits).

The reason this is equivalent to Definition 10.1 (which requires a larger value of m in general) is that each such M_j can be expressed as a linear combination of Pauli operations on k or fewer qubits. When tensored with the identity operation on the remaining qubits, we obtain a linear combination of Pauli operations having weight at most k. It should be mentioned that Definition 10.1 not only offers a slightly simpler formal definition, but is also convenient for both the problem statement that follows as well as the proof that this problem is contained in QMA.

For an example of a local Hamiltonian, consider a 3CNF formula ϕ having clauses $C_1, \ldots, C_m$ in Boolean variables $X_1, \ldots, X_n$. For each clause C_j, let Q_j be the projection onto the span of the standard basis states $|z\rangle$, for $z \in \{0,1\}^n$, such that the assignment z violates C_j. For instance,

$$C = X_1 \vee \overline{X_2} \vee X_3 \;\longrightarrow\; Q = |010\rangle\langle 010| \otimes \mathbb{1}^{\otimes(n-3)}. \tag{10.14}$$

Each Q_j acts non-trivially on just 3 qubits, so

$$H = Q_1 + \cdots + Q_m \tag{10.15}$$

is a 3-local Hamiltonian. This Hamiltonian happens to be relatively simple because it is diagonal: $H|z\rangle$ is equal to $|z\rangle$ times the number of clauses of ϕ that z violates, for every standard basis state $|z\rangle$. So, we can think of each term Q_j as *penalizing* the assignments that violate the clause C_j. The smallest eigenvalue (or *ground state energy*) $\lambda_{\min}(H)$ is equal to the minimum, over all assignments to the variables, of the number of clauses violated. In particular, $\lambda_{\min}(H) = 0$ if ϕ is satisfiable and $\lambda_{\min}(H) \geq 1$ otherwise.

Problem statement

Now we are prepared to state the k-local Hamiltonian problem itself.

k-LOCAL HAMILTONIAN (k-LH)

Input	Pauli operations $P_1, \ldots, P_m \in \mathcal{P}_n$ on n qubits, each having weight at most k, expressed as strings in $\{\mathbb{1}, X, Y, Z\}^n$; rational number coefficients $\gamma_1, \ldots, \gamma_m \in [-1, 1]$; and a positive integer K written in unary.
Yes	$\lambda_{\min}(\gamma_1 P_1 + \cdots + \gamma_m P_m) \leq -1/K$.
No	$\lambda_{\min}(\gamma_1 P_1 + \cdots + \gamma_m P_m) \geq 1/K$.

Here, $\lambda_{\min}(H)$ refers to the minimum eigenvalue of H, which is given by

$$\lambda_{\min}(H) = \min_{\rho} \operatorname{Tr}(H\rho), \tag{10.16}$$

where the minimum is over all density matrices ρ of n qubits. The minimum is always attained for a pure state, $\rho = |\psi\rangle\langle\psi|$, where $|\psi\rangle$ is a unit eigenvector of H having eigenvalue $\lambda_{\min}(H)$.

Thus, the input describes a k-local Hamiltonian

$$H = \gamma_1 P_1 + \cdots + \gamma_m P_m, \tag{10.17}$$

and the goal is to determine whether the smallest eigenvalue of H is at most $-1/K$ or at least $1/K$, promised that one of these two possibilities is true.

Note that by requiring K to be input in unary, together with the requirement that each coefficient lies in $[-1, 1]$, the promise guarantees that the gap between the values $-1/K$ and $1/K$ is nonnegligible relative to the norms of the terms of the Hamiltonian. It is common that $-1/K$ and $1/K$ are replaced by arbitrary rational numbers α and β, necessarily separated by an inverse polynomial gap but not necessarily centered around 0, but no generality is lost in taking $-1/K$ and $1/K$ as in the problem statement.

10.3 The local Hamiltonian problem is in QMA

Next we will prove that the k-local Hamiltonian problem is contained in QMA for every positive integer constant k.

The verification procedure

For a given input $P_1, \ldots, P_m \in \mathcal{P}_n$, $\gamma_1, \ldots, \gamma_m \in [-1, 1]$, and K as in the problem statement, we may assume that the coefficients $\gamma_1, \ldots, \gamma_m$ are all nonzero, as terms with zero coefficients contribute nothing to the Hamiltonian and can simply be ignored. Consider the following procedure, to be performed on an n-qubit register W:

1. Compute
$$\Gamma = |\gamma_1| + \cdots + |\gamma_m| \tag{10.18}$$
(which is necessarily positive for any input satisfying the promise) and define a probability distribution μ on $\{1, \ldots, m\}$ by
$$\mu(j) = \frac{|\gamma_j|}{\Gamma}. \tag{10.19}$$

2. Select an index $j \in \{1, \ldots, m\}$ with probability $\mu(j)$ and perform the Pauli measurement P_j on W, obtaining an outcome $a \in \{+1, -1\}$.

3. Accept if $a\gamma_j < 0$ (i.e., a and γ_j have opposite sign) and reject otherwise.

Analysis of the procedure

When the verification procedure described above is performed when W is in the state ρ, it accepts with probability
$$\frac{1}{2} - \frac{\mathrm{Tr}(H\rho)}{2\Gamma}. \tag{10.20}$$

To see this, let us define λ_j to be the probability the procedure accepts conditioned on $j \in \{1, \ldots, m\}$ being selected in step 2. The condition for acceptance is that $a\gamma_j < 0$, which is equivalent to a being opposite to the sign of γ_j. Thus, considering the formula (10.7), we have
$$\lambda_j = \begin{cases} \big(1 - \mathrm{Tr}(P_j\rho)\big)/2 & \text{if } \gamma_j > 0 \\ \big(1 + \mathrm{Tr}(P_j\rho)\big)/2 & \text{if } \gamma_j < 0. \end{cases} \tag{10.21}$$

Equivalently,
$$\lambda_j = \frac{1}{2} - \frac{\gamma_j \,\mathrm{Tr}(P_j\rho)}{2|\gamma_j|}. \tag{10.22}$$

Averaging according to the probabilities for choosing each j reveals that the overall probability of acceptance is

$$\sum_{j=1}^{m} \mu(j)\,\lambda_j = \sum_{j=1}^{m} \frac{|\gamma_j|}{\Gamma}\left(\frac{1}{2} - \frac{\gamma_j \operatorname{Tr}(P_j\rho)}{2|\gamma_j|}\right) \tag{10.23}$$

$$= \frac{1}{2} - \frac{1}{2\Gamma}\sum_{j=1}^{m} \gamma_j \operatorname{Tr}(P_j\rho) \tag{10.24}$$

$$= \frac{1}{2} - \frac{\operatorname{Tr}(H\rho)}{2\Gamma}, \tag{10.25}$$

as claimed.

Sampling with uniformly random bits

Formally speaking, it may not be possible to sample exactly from the distribution μ defined in the verification procedure using uniformly generated random bits, but it suffices to do this approximately.

One simple way to proceed starts by expressing each $|\gamma_j|$ as

$$|\gamma_j| = \frac{b_j}{D} \tag{10.26}$$

for a positive integer b_j and a common denominator D. We can then round the sum of the numerators up to the nearest power of 2, which is 2^r for

$$r = \left\lceil \log_2(b_1 + \cdots + b_m) \right\rceil, \tag{10.27}$$

choose a number uniformly from the set $\{0, \ldots, 2^r - 1\}$ using r uniformly generated random bits, and select the outcome $j \in \{1, \ldots, m\}$ in case the random number falls in the range

$$\left\{ \sum_{\ell=1}^{j-1} b_\ell, \ldots, \sum_{\ell=1}^{j} b_\ell - 1 \right\}. \tag{10.28}$$

If the random number is at least the sum $b_1 + \cdots + b_m$ then the random selection fails — which can happen with probability at most $1/2$. Conditioned on the selection not failing, the sample obtained is selected from the distribution μ.

This procedure can be repeated up to a polynomial number of times until success, giving an exponentially small probability of failure. Alternatively, even a failure probability of $1/2$ can be tolerated for the purposes of showing that the local Hamiltonian problem is in QMA, as will be argued shortly.

An alternative is to round each $|\gamma_j|$ to the form (10.26) where D is a power of 2 and $b_1 + \cdots + b_m = D$ and to analyze the error incurred in doing this. This is arguably a more practical approach, but not quite as simple to analyze, so we will not pursue this option further in these notes.

Membership in QMA

We now have the pieces required to prove that the local Hamiltonian problem is in QMA.

Theorem 10.2. *For every positive integer k, k-LH $\in$ QMA.*

Proof. Assume k is fixed for the entire proof.

We will define a family $\{Q_x : x \in \Sigma^*\}$ of quantum circuits, each having $|x|$ input qubits and one output qubit, that implements the verification procedure described in the subsection above. These circuits will ignore all but the first n of these qubits, where n depends on the instance of k-LH. (So n does not denote the length of x in the context of this proof.) These n qubits play the role of the register W in the verification procedure.

If x does not encode an input to k-LH having at least one nonzero coefficient, we will take Q_x to be an arbitrary quantum circuit having $|x|$ input qubits and one output qubit (e.g., $|x|$ erasure gates on the inputs along with one ancillary qubit that produces the output). This choice has no bearing on the remainder of the proof, but only guarantees that $\{Q_x : x \in \Sigma^*\}$ has the proper form.

For the remainder of the proof we will focus on the case that x encodes an instance of k-LH having at least one non-zero coefficient. Note that this implies that $|x| \geq n$, which guarantees that Q_x has at least n input qubits.

A polynomial-time DTM M that computes Q_x from x first discards any terms having zero coefficients, then computes Γ and μ as described in the verification procedure, and then constructs a circuit that takes no input and attempts to output a random choice of $j \in \{1, \ldots, m\}$ according to the distribution μ, succeeding with probability at least $1/2$ as described above. This may be done by using the methods described in the proof of Proposition 5.4 for proving the containment BPP $\subseteq$ BQP. The DTM M then composes this circuit with one that performs the Pauli measurement P_j on the first n input qubits of Q_x using phase estimation, again as described above.

Finally, M composes this circuit with one that accepts or rejects as described in step 3 of the verification procedure. In the case that the random selection of j fails, the circuit representing this final step of the procedure accepts or rejects

uniformly at random. The resulting circuit is Q_x, and it is routine to verify that the computation $x \mapsto \langle Q_x \rangle$ just described can be computed in polynomial time.

Suppose now that x encodes an input of k-LH satisfying the promise, and let $H = \gamma_1 P_1 + \cdots + \gamma_m P_m$. The promise implies that $\lambda_{\min}(H) \neq 0$, so at least one of the coefficients $\gamma_1, \ldots, \gamma_m$ is nonzero, so Q_x is a circuit implementing the verification procedure as just described. Let δ denote the probability that the random selection of j fails, which depends on x but always satisfies $\delta \leq 1/2$.

Consider any state σ of $|x|$ qubits, and let ρ be the n-qubit state obtained by tracing out all but the first n qubits of σ. Conditioned on the random selection not failing, the index j is chosen according to the distribution μ, and in this case Q_x performs steps 2 and 3 of the verification procedure on ρ. As the analysis presented previously revealed, it then accepts with the probability given by (10.20), which is replicated here for convenience.

$$\frac{1}{2} - \frac{\mathrm{Tr}(H\rho)}{2\Gamma} \tag{10.20}$$

Conditioned on the selection failing, on the other hand, it accepts with probability $1/2$. It follows that

$$\Pr(Q_x \text{ accepts } \sigma) = \frac{\delta}{2} + (1-\delta)\left(\frac{1}{2} - \frac{\mathrm{Tr}(H\rho)}{2\Gamma}\right) = \frac{1}{2} - (1-\delta)\,\frac{\mathrm{Tr}(H\rho)}{2\Gamma}. \tag{10.29}$$

If x is a yes-input, then (10.16) implies that there exists a state ρ of n qubits satisfying $\mathrm{Tr}(H\rho) \leq -1/K$. Taking σ to be ρ tensored with any state of the remaining $|x| - n$ qubits, we find that

$$\Pr(Q_x \text{ accepts } \sigma) \geq \frac{1}{2} + \frac{1-\delta}{2\Gamma K} \geq \frac{1}{2} + \frac{1}{4\Gamma K}. \tag{10.30}$$

If instead x is a no-input, then every state ρ of n qubits satisfies $\mathrm{Tr}(H\rho) \geq 1/K$, again by (10.16), and therefore every state σ of $|x|$ qubits satisfies

$$\Pr(Q_x \text{ accepts } \sigma) \leq \frac{1}{2} - \frac{1-\delta}{2\Gamma K} \leq \frac{1}{2} - \frac{1}{4\Gamma K}. \tag{10.31}$$

It remains to observe that the gap between these two probability bounds is at least inverse-polynomial in the input length. It suffices to prove an upper bound on ΓK in terms of $|x|$. Each coefficient satisfies $|\gamma_j| \leq 1$, so $\Gamma \leq m$. Moreover, each of the m Pauli operations requires at least one symbol of x to describe it, and K is written in unary, so $m \leq |x|$ and $K \leq |x|$. Consequently, $\Gamma K \leq |x|^2$, and therefore

$$\frac{1}{4\Gamma K} \geq \frac{1}{4|x|^2}. \tag{10.32}$$

The proof is completed by choosing the functions α and β appropriately, so that $\alpha(|x|) - \beta(|x|) \geq 1/(2|x|^2)$, and then applying Theorem 9.4. □

10.4 QMA-hardness of the 5-local Hamiltonian problem

Now we will prove that the k-local Hamiltonian problem is QMA-hard, with respect to Karp reductions, for $k = 5$. This is done in the subsections that follow.

A normal form for verification circuits

To reduce an arbitrary promise problem in QMA to an instance of 5-LH requires the construction of a Hamiltonian for each quantum circuit in a polynomial-time uniform family for the promise problem, according to the definition of QMA (Definition 8.9). The first step is to prove that these quantum circuits can have a certain *normal form* that facilitates this construction.

To begin, we will require circuits consisting of only one- and two-qubit gates, which disallows Toffoli gates. (The proof would go through without complications if we did allow Toffoli gates, but we would end up with an instance of 6-LH rather than 5-LH.) It is also convenient to use gates having Gaussian rational entries, which disallows Hadamard gates. A simple remedy for both issues is to change the gate set from $\{H, S, \mathrm{TOF}\}$ to $\{H \otimes H, S, \Lambda(S)\}$, where $\Lambda(S)$ is a controlled-S gate. The matrices corresponding to $\Lambda(S)$ and $H \otimes H$ are as follows:

$$\Lambda(S) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{pmatrix} \quad \text{and} \quad H \otimes H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{10.33}$$

We note that Toffoli gates can be implemented using H, $\Lambda(S)$, $\Lambda(S^*)$, and CNOT gates as shown in Figure 10.2; CNOT gates can be implemented using two H gates and two $\Lambda(S)$ gates; and $\Lambda(S^*) = \Lambda(S)^3$. Finally, an H gate is easily implemented using an $H \otimes H$ gate by taking one of the two qubits to be a dummy qubit added to the circuit that will not affect its computation. Thus, it is routine to convert a given quantum circuit composed of H, S, and TOF gates to an equivalent one using $H \otimes H$, S, and $\Lambda(S)$ gates.

In addition, it will be convenient to assume that the number of gates in our circuits is determined by a polynomial resource bound (therefore depending only on the input length).

The following lemma establishes that all of these requirements can indeed be met. In it, extending the notation of Chapter 8 to unitary circuits as in Section 9.2,

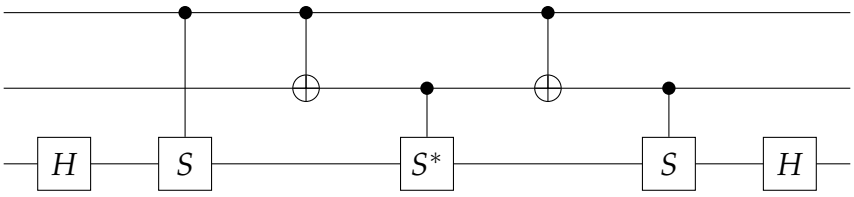


Figure 10.2: An implementation of a Toffoli gate as a product of Hadamard, CNOT, controlled-S, and controlled-S^* gates.

we write

$$\Pr(U \text{ accepts } |\psi\rangle) = \left\| \Pi_1 U \left(|\psi\rangle \otimes |0^q\rangle \right) \right\|^2 \tag{10.34}$$

for $\Pi_1 = |1\rangle\langle 1| \otimes \mathbb{1}^{\otimes(p+q-1)}$, where U is a unitary quantum circuit acting on $p+q$ qubits. The first p qubits U acts on form a witness register and the remaining q qubits are ancillary qubits. The first qubit is taken to be the output qubit after U is performed.

Lemma 10.3. *Let $A = (A_{\text{yes}}, A_{\text{no}})$ be a promise problem in* QMA. *There exist polynomial resource bounds p, q, and $s \geq 2$, along with a polynomial-time uniform family*

$$\{U_x \,:\, x \in \Sigma^*\} \tag{10.35}$$

of unitary quantum circuits, where each U_x acts on $p+q$ qubits and is composed of exactly s gates, each drawn from the set

$$\{S,\ \Lambda(S),\ H \otimes H\}, \tag{10.36}$$

such that these two implications are true:

1. *If $x \in A_{\text{yes}}$, then there exists a p-qubit quantum state vector $|\psi\rangle$ such that*

$$\Pr(U_x \text{ accepts } |\psi\rangle) \geq 1 - \frac{1}{256(s+1)}. \tag{10.37}$$

2. *If $x \in A_{\text{no}}$, then every p-qubit quantum state vector $|\psi\rangle$ satisfies*

$$\Pr(U_x \text{ accepts } |\psi\rangle) \leq \frac{1}{4}. \tag{10.38}$$

Proof. By Theorem 9.4, the assumption $A \in$ QMA implies that there exists a polynomial resource bound p such that, for any chosen polynomial resource bound r, we have $A \in \text{QMA}_p(1 - 2^{-r}, 2^{-r})$. We will choose a specific r at the end of the

proof. Let $\{Q_x : x \in \Sigma^*\}$ be the polynomial-time uniform family of quantum circuits (over our standard gate set) constructed in the proof of Theorem 9.4, which meets the requirements of the definition of $\mathrm{QMA}_p(1 - 2^{-r}, 2^{-r})$.

Next, let $\{V_x : x \in \Sigma^*\}$ be a family of unitary quantum circuits over the gate set $\{H, S, \mathrm{TOF}\}$ dilating the circuits $\{Q_x : x \in \Sigma^*\}$ as in Proposition 5.2. Specifically, we will assume that each V_x uses q_0 ancillary qubits and has s_0 gates, for polynomial resource bounds q_0 and s_0. As each V_x dilates Q_x, we have (as in Section 9.2)

$$\langle 1|Q_x(|\psi\rangle\langle\psi|)|1\rangle = \Pr(V_x \text{ accepts } |\psi\rangle) \tag{10.39}$$

for every p-qubit state $|\psi\rangle$.

We will transform each V_x into U_x satisfying the requirements of the lemma in three steps, none of which changes $\Pr(V_x \text{ accepts } |\psi\rangle)$.

1. Each Toffoli gate is simulated by H and $\Lambda(S)$ gates as described previously. This requires 15 gates in total, so the circuit constructed from V_x in this way has at most $15s_0$ gates.

2. One new ancillary qubit is added, and each Hadamard gate is replaced by $H \otimes H$ acting on its qubit and this new one. The circuit constructed from V_x therefore implements either $V_x \otimes \mathbb{1}$ or $V_x \otimes H$, depending on whether V_x uses an even or odd number of Hadamard gates, and uses $q = q_0 + 1$ ancillary qubits.

3. Let $s = 15s_0 + 2$ and let U_x be the circuit obtained from V_x by performing the two previous steps along with prepending enough S gates to the new ancillary qubit so that exactly s gates are used.

Each step is a simple, local substitution, so the family $\{U_x : x \in \Sigma^*\}$ is polynomial-time uniform.

It remains to choose r so that $2^{-r} \leq 1/(256(s+1))$, which also gives $2^{-r} \leq 1/4$, for then the implications (10.37) and (10.38) follow from $A \in \mathrm{QMA}_p(1 - 2^{-r}, 2^{-r})$. This step does require care because s depends on r. The circuits of Construction 9.3 consist of a number of copies of a fixed dilation and its inverse that is linear in r, together with a threshold computation on $O(r)$ bits. It follows that $s \leq r^d s_1$ for a constant d and a polynomial resource bound s_1, neither of which depends on r. Taking $r(n) = s_1(n) + c$ for a sufficiently large constant c satisfies the requirements on r. □

Hamiltonian construction

For a given unitary quantum circuit $U = U_s \cdots U_1$ acting on $p + q$ qubits and having the form guaranteed by Lemma 10.3, our ultimate goal is to define a 5-local Hamiltonian whose minimum eigenvalue is small when there exists a p-qubit state $|\psi\rangle$ that U accepts with high probability and large when U accepts every p-qubit state $|\psi\rangle$ with small probability.

To accomplish this, we will introduce an s-qubit *clock register* $\mathsf{C} = (\mathsf{C}_1, \ldots, \mathsf{C}_s)$, which will store time steps of U, corresponding to $t \in \{0, \ldots, s\}$, where each t is encoded as the standard basis state $|1^t 0^{s-t}\rangle$. This particular encoding is important for the sake of locality. In particular, each string $1^t 0^{s-t}$ uniquely determines t through conditions on one or two bits: if the first bit is 0, then $t = 0$; if the last bit is 1, then $t = s$; and otherwise t is determined by the unique starting position of the substring 10. For such a clock register C, together with the registers W and Y of U, we introduce the *history state* of U on a p-qubit state $|\psi\rangle$ as

$$|\phi\rangle = \frac{1}{\sqrt{s+1}} \sum_{t=0}^{s} |1^t 0^{s-t}\rangle \otimes U_t \cdots U_1 \big(|\psi\rangle \otimes |0^q\rangle\big). \tag{10.40}$$

The fundamental idea behind the Hamiltonian construction is that 1) history states of U leading to acceptance will have low energy, and 2) the ground state energy is high in the absence of a witness state $|\psi\rangle$ leading U to acceptance.

In analogy with the example of Section 10.2, we say that a term T of the Hamiltonian *penalizes* a state $|\chi\rangle$ by the amount $\langle\chi|T|\chi\rangle$ that it contributes to the energy $\langle\chi|H|\chi\rangle$. Each of the terms defined below is a projection, so this amount lies between 0 and 1.

The first step is to define a portion of the Hamiltonian that effectively penalizes illegal states of the clock register.

$$H_{\text{clock}} = \sum_{t=1}^{s-1} |01\rangle\langle 01|_{\mathsf{C}_t \mathsf{C}_{t+1}}. \tag{10.41}$$

Here, the subscript $\mathsf{C}_t\mathsf{C}_{t+1}$ indicates an action on the registers C_t and C_{t+1}, in that order. The understanding is that this action is tensored with the identity operation on all other qubits. This notation will be used more generally for other choices of qubits. The Hamiltonian H_{clock} is 2-local.

Next, we define another portion of the Hamiltonian that penalizes invalid starting states as well as rejecting final states. The assumption that the computation of U requires Y to be initialized to $|0^q\rangle$ is reflected by defining

$$H_{\text{in}} = \sum_{i=1}^{q} |0\rangle\langle 0|_{\mathsf{C}_1} \otimes |1\rangle\langle 1|_{\mathsf{Y}_i}. \tag{10.42}$$

The only way C_1 is in the state $|0\rangle$ for a legal state of the clock register is if $t = 0$, in which case $|1\rangle$ is penalized for every qubit of Y. Along similar lines, we define

$$H_{\text{out}} = |1\rangle\langle 1|_{\mathsf{C}_s} \otimes |0\rangle\langle 0|_{\mathsf{W}_1}. \tag{10.43}$$

This Hamiltonian penalizes rejecting states when $t = s$. Both H_{in} and H_{out} are 2-local.

Finally, we define Hamiltonian terms whose low-energy states represent valid computations according to $U_1, \ldots, U_s$. There are s of them in total:

$$\begin{aligned} H_1 &= |00\rangle\langle 00|_{\mathsf{C}_1\mathsf{C}_2} + |10\rangle\langle 10|_{\mathsf{C}_1\mathsf{C}_2} \\ &\quad - |10\rangle\langle 00|_{\mathsf{C}_1\mathsf{C}_2} \otimes U_1 - |00\rangle\langle 10|_{\mathsf{C}_1\mathsf{C}_2} \otimes U_1^*, \end{aligned} \tag{10.44}$$

$$\begin{aligned} H_t &= |100\rangle\langle 100|_{\mathsf{C}_{t-1}\mathsf{C}_t\mathsf{C}_{t+1}} + |110\rangle\langle 110|_{\mathsf{C}_{t-1}\mathsf{C}_t\mathsf{C}_{t+1}} \\ &\quad - |110\rangle\langle 100|_{\mathsf{C}_{t-1}\mathsf{C}_t\mathsf{C}_{t+1}} \otimes U_t - |100\rangle\langle 110|_{\mathsf{C}_{t-1}\mathsf{C}_t\mathsf{C}_{t+1}} \otimes U_t^*, \end{aligned} \tag{10.45}$$

$$\begin{aligned} H_s &= |10\rangle\langle 10|_{\mathsf{C}_{s-1}\mathsf{C}_s} + |11\rangle\langle 11|_{\mathsf{C}_{s-1}\mathsf{C}_s} \\ &\quad - |11\rangle\langle 10|_{\mathsf{C}_{s-1}\mathsf{C}_s} \otimes U_s - |10\rangle\langle 11|_{\mathsf{C}_{s-1}\mathsf{C}_s} \otimes U_s^*, \end{aligned} \tag{10.46}$$

for every $t \in \{2, \ldots, s-1\}$ in (10.45). The locality of these Hamiltonian terms depends on $U_1, \ldots, U_s$ — but if U_t acts on 2 or fewer qubits, then H_t is 5-local.

The final Hamiltonian is a weighted sum of the Hamiltonians just defined.

$$H = H_{\text{clock}} + H_{\text{in}} + \frac{1}{2}\sum_{t=1}^{s} H_t + H_{\text{out}}. \tag{10.47}$$

This is sometimes called the *Feynman–Kitaev Hamiltonian* of U, or just the Hamiltonian corresponding to U for brevity. Each term of H is 5-local, and therefore H is 5-local.

The Hamiltonian H is a sum of $2s + q$ terms: the $s - 1$ terms of H_{clock}, the q terms of H_{in}, the s terms $\frac{1}{2}H_t$, and H_{out}. Each of these terms acts on at most 5 qubits, and each is a projection. For the terms of H_{clock}, H_{in}, and H_{out} this is evident, as each is a tensor product of projections. Each H_t is evidently Hermitian, and squaring the definitions reveals that

$$H_t^2 = 2H_t, \tag{10.48}$$

so $\frac{1}{2}H_t$ is a projection. In particular, every term of H is positive semidefinite.

Finally, we note that when every gate U_t is drawn from the set (10.36), every entry of every term of H is contained in the set

$$\{0, 1, \pm\tfrac{1}{4}, \pm\tfrac{1}{2}, \pm\tfrac{i}{2}\}. \tag{10.49}$$

Legal and illegal clock strings

Now let us study the clock register and its relation to the Hamiltonian more closely. Assume the circuit $U = U_s \cdots U_1$ acting on $p+q$ qubits is fixed for this discussion.

Define the set of *legal clock strings* as

$$L = \left\{1^t 0^{s-t} \,:\, t \in \{0, \ldots, s\}\right\}, \tag{10.50}$$

which can alternatively be characterized as

$$L = \left\{y \in \{0,1\}^s \,:\, 01 \text{ is not a substring of } y\right\}. \tag{10.51}$$

Also define the projection

$$\Pi_{\mathcal{L}} = \sum_{y \in L} |y\rangle\langle y| \otimes \mathbb{1}, \tag{10.52}$$

where $\mathbb{1}$ denotes the identity on $p+q$ qubits, and let $\mathcal{L}$ be the image of $\Pi_{\mathcal{L}}$. This subspace will be called the *legal subspace* of the $s+p+q$ qubits corresponding to the registers $(\mathsf{C}, \mathsf{W}, \mathsf{Y})$. This subspace contains the history state (10.40) for every p-qubit state $|\psi\rangle$. We will later focus on the action of H on $\mathcal{L}$, and to this end we must understand what happens on $\mathcal{L}^\perp$, the orthogonal complement of $\mathcal{L}$.

For an arbitrary binary string $y \in \{0,1\}^s$ and any $(p+q)$-qubit vector $|\eta\rangle$, observe that

$$H_{\text{clock}}(|y\rangle \otimes |\eta\rangle) = |y|_{01} |y\rangle \otimes |\eta\rangle, \tag{10.53}$$

where $|y|_{01}$ is the number of occurrences of the substring 01 in y. In particular, H_{clock} is diagonal in the standard basis (as is $\Pi_{\mathcal{L}}$), and

$$H_{\text{clock}} \Pi_{\mathcal{L}} = 0 \tag{10.54}$$

because $|y|_{01} = 0$ for $y \in L$. This implies that H_{clock} annihilates history states, and therefore gives them no penalty.

We can also observe from (10.53) that H_{clock} gives a significant penalty to vectors in $\mathcal{L}^\perp$. In particular,

$$\mathbb{1} - \Pi_{\mathcal{L}} \leq H_{\text{clock}} \tag{10.55}$$

because $|y|_{01} \geq 1$ for $y \notin L$. This inequality will effectively allow us to disregard the subspace $\mathcal{L}^\perp$ when considering low energy states of H. It is enough to observe that, for every vector $|\chi\rangle \in \mathcal{L}^\perp$, we have

$$\langle\chi| H_{\text{clock}} |\chi\rangle \geq \langle\chi|\chi\rangle. \tag{10.56}$$

Finally, we will show that the Hamiltonian H respects the decomposition of the entire $s+p+q$ qubit space it acts upon into the direct sum of $\mathcal{L}$ and $\mathcal{L}^\perp$. In fact, its

action within $\mathcal{L}^{\perp}$ can be decomposed further, with one subspace for each positive integer number of occurrences of the substring 01 in the standard basis states of the clock register.

In greater detail, we may view H as a $2^s \times 2^s$ block matrix, whose block with coordinates (y, z) is the $2^{p+q} \times 2^{p+q}$ matrix

$$B_{y,z} = \big(\langle y| \otimes \mathbb{1}\big) H \big(|z\rangle \otimes \mathbb{1}\big), \tag{10.57}$$

so that

$$H = \sum_{y,z \in \{0,1\}^s} |y\rangle\langle z| \otimes B_{y,z}. \tag{10.58}$$

It is possible to do this with any matrix, and we will better understand the structure of H by writing it in this way. In particular, we will argue that

$$B_{y,z} = 0 \quad \text{whenever} \quad |y|_{01} \neq |z|_{01}. \tag{10.59}$$

Intuitively speaking, this implies that the Hamiltonian never changes the number of occurrences of the substring 01 in any standard basis state of the clock register.

To see this, we can use linearity: each $B_{y,z}$ is given by a sum over corresponding matrices for the individual terms of H. In fact, for every term in the Hamiltonian that matrix is zero whenever y and z have differing numbers of occurrences of the substring 01.

1. For H_{in}, H_{out}, and H_{clock} this is immediate from the fact that these matrices are diagonal in the standard basis, so there cannot be any non-zero off-diagonal blocks at all.

2. For H_1, we again have contributions to the diagonal blocks that we can ignore, as well as non-zero contributions to the off-diagonal blocks that come from the portion of H_1 given by

$$-|10\rangle\langle 00|_{\mathsf{C}_1\mathsf{C}_2} \otimes U_1 - |00\rangle\langle 10|_{\mathsf{C}_1\mathsf{C}_2} \otimes U_1^*. \tag{10.60}$$

   Observing that

$$|00u|_{01} = |10u|_{01} \tag{10.61}$$

   for every binary string u, we find that

$$(\langle y| \otimes \mathbb{1}) H_1 (|z\rangle \otimes \mathbb{1}) = 0 \tag{10.62}$$

   whenever $|y|_{01} \neq |z|_{01}$.

3. For H_t with $1 < t < s$, the situation is similar, except that the contributions to the off-diagonal blocks come from

$$-|110\rangle\langle 100|_{\mathsf{C}_{t-1}\mathsf{C}_t\mathsf{C}_{t+1}} \otimes U_t - |100\rangle\langle 110|_{\mathsf{C}_{t-1}\mathsf{C}_t\mathsf{C}_{t+1}} \otimes U_t^*. \tag{10.63}$$

   This time we make use of the equality

$$|u100v|_{01} = |u110v|_{01} \tag{10.64}$$

   for all binary strings u and v to obtain

$$(\langle y| \otimes \mathbb{1})H_t(|z\rangle \otimes \mathbb{1}) = 0 \tag{10.65}$$

   whenever $|y|_{01} \neq |z|_{01}$.

4. Finally, for H_s, the contributions to the off-diagonal blocks come from

$$-|11\rangle\langle 10|_{\mathsf{C}_{s-1}\mathsf{C}_s} \otimes U_s - |10\rangle\langle 11|_{\mathsf{C}_{s-1}\mathsf{C}_s} \otimes U_s^*. \tag{10.66}$$

   Here we use the equality

$$|u10|_{01} = |u11|_{01}, \tag{10.67}$$

   which holds for every binary string u, to determine that

$$(\langle y| \otimes \mathbb{1})H_s(|z\rangle \otimes \mathbb{1}) = 0 \tag{10.68}$$

   whenever $|y|_{01} \neq |z|_{01}$.

For the purposes of this chapter, the important implication we draw from this analysis is that H and $\Pi_{\mathcal{L}}$ commute. In particular, this implies that

$$\Pi_{\mathcal{L}} H\big(\mathbb{1} - \Pi_{\mathcal{L}}\big) = 0 = \big(\mathbb{1} - \Pi_{\mathcal{L}}\big) H \Pi_{\mathcal{L}}. \tag{10.69}$$

Hamiltonian action on the legal subspace

Now let us record how each term of H acts on vectors in the legal subspace, which is where most of the analysis that follows takes place. For the sake of clarity, we will write $|t\rangle$ in place of $|1^t 0^{s-t}\rangle$ for each $t \in \{0, \ldots, s\}$. Thus, every vector $|\chi\rangle \in \mathcal{L}$ can be expressed as

$$|\chi\rangle = \sum_{t=0}^{s} |t\rangle \otimes |\chi_t\rangle \tag{10.70}$$

for vectors $|\chi_0\rangle, \ldots, |\chi_s\rangle$ of (W, Y). Note that these are generally not unit vectors: we have

$$\||\chi\rangle\|^2 = \||\chi_0\rangle\|^2 + \cdots + \||\chi_s\rangle\|^2. \tag{10.71}$$

Next, define the following projections, all acting on (W, Y):

$$\Pi_0 = |0\rangle\langle 0| \otimes \mathbb{1}^{\otimes(p+q-1)}, \tag{10.72}$$

$$\Pi_1 = |1\rangle\langle 1| \otimes \mathbb{1}^{\otimes(p+q-1)}, \tag{10.73}$$

$$\Delta_0 = \mathbb{1}^{\otimes p} \otimes |0^q\rangle\langle 0^q|, \tag{10.74}$$

$$\Delta_1 = \mathbb{1}^{\otimes p} \otimes \left(\mathbb{1}^{\otimes q} - |0^q\rangle\langle 0^q|\right). \tag{10.75}$$

These projections are similar to ones defined in the proof of Theorem 9.4 (though Π_0 and Π_1 are not conjugated by U^* this time), and play a similar role: $\{\Pi_0, \Pi_1\}$ describes a standard basis measurement of the output qubit while $\{\Delta_0, \Delta_1\}$ describes a measurement determining if the ancillary qubits are initialized or not.

Using the notation now established, we have

$$\langle\chi|H_{\text{clock}}|\chi\rangle = 0, \tag{10.76}$$

$$\langle\chi|H_{\text{in}}|\chi\rangle \geq \left\|\Delta_1|\chi_0\rangle\right\|^2, \tag{10.77}$$

$$\langle\chi|H_{\text{out}}|\chi\rangle = \left\|\Pi_0|\chi_s\rangle\right\|^2, \tag{10.78}$$

$$\langle\chi|H_t|\chi\rangle = \left\|U_t|\chi_{t-1}\rangle - |\chi_t\rangle\right\|^2 \tag{10.79}$$

for $t = 1, \ldots, s$. Each is established by matching legal clock strings to the first tensor factor in the Hamiltonian terms, using the observation that

$$\sum_{i=1}^{q} |1\rangle\langle 1|_{\mathsf{Y}_i} \geq \Delta_1 \tag{10.80}$$

in (10.77). For (10.79), one may use the equality

$$\left\||\chi_{t-1}\rangle\right\|^2 + \left\||\chi_t\rangle\right\|^2 - \langle\chi_t|U_t|\chi_{t-1}\rangle - \langle\chi_{t-1}|U_t^*|\chi_t\rangle = \left\|U_t|\chi_{t-1}\rangle - |\chi_t\rangle\right\|^2, \tag{10.81}$$

which holds whenever U_t is unitary.

Completeness

Now suppose that there exists a p-qubit state vector $|\psi\rangle$ that U accepts with high probability:

$$\Pr(U \text{ accepts } |\psi\rangle) \geq 1 - \varepsilon \tag{10.82}$$

for $\varepsilon \in [0, 1]$. We will prove that

$$\lambda_{\min}(H) \leq \frac{\varepsilon}{s+1} \tag{10.83}$$

by showing that the history state $|\phi\rangle$ of U on $|\psi\rangle$, defined in (10.40), satisfies $\langle\phi|H|\phi\rangle \leq \varepsilon/(s+1)$. The history state is replicated here for convenience.

$$|\phi\rangle = \frac{1}{\sqrt{s+1}} \sum_{t=0}^{s} |1^t 0^{s-t}\rangle \otimes U_t \cdots U_1 \big(|\psi\rangle \otimes |0^q\rangle\big) \tag{10.40}$$

This is a unit vector, and an inspection of the Hamiltonian terms other than H_{out} reveals that

$$\langle\phi|H_{\text{clock}}|\phi\rangle = \langle\phi|H_{\text{in}}|\phi\rangle = \langle\phi|H_t|\phi\rangle = 0 \tag{10.84}$$

for every $t \in \{1, \ldots, s\}$. Finally, by (10.78),

$$\langle\phi|H_{\text{out}}|\phi\rangle = \frac{\big\|\Pi_0 U(|\psi\rangle \otimes |0^q\rangle)\big\|^2}{s+1} = \frac{1 - \Pr(U \text{ accepts } |\psi\rangle)}{s+1} \leq \frac{\varepsilon}{s+1}, \tag{10.85}$$

and therefore

$$\lambda_{\min}(H) \leq \langle\phi|H|\phi\rangle \leq \frac{\varepsilon}{s+1}, \tag{10.86}$$

as claimed.

Soundness

Now suppose that U accepts every p-qubit state vector with low probability:

$$\Pr(U \text{ accepts } |\psi\rangle) \leq \frac{1}{4} \tag{10.87}$$

for every p-qubit state vector $|\psi\rangle$. This is equivalent to

$$\big\|\Pi_1 U\big(|\xi\rangle \otimes |0^q\rangle\big)\big\| \leq \frac{1}{2}\big\||\xi\rangle\big\| \tag{10.88}$$

for every vector $|\xi\rangle$ of p qubits. Our goal will be to prove that

$$\lambda_{\min}(H) \geq \frac{1}{128(s+1)^2}. \tag{10.89}$$

We will first consider vectors in the legal subspace. Let $|\chi\rangle \in \mathcal{L}$ be expressed as in (10.70), and suppose $\delta > 0$ is a real number satisfying

$$\langle\chi|H_{\text{in}}|\chi\rangle \leq \delta^2, \quad \langle\chi|H_{\text{out}}|\chi\rangle \leq \delta^2, \quad \text{and} \quad \sum_{t=1}^{s} \langle\chi|H_t|\chi\rangle \leq \delta^2. \tag{10.90}$$

By (10.77)–(10.79), it follows that

$$\big\|\Delta_1|\chi_0\rangle\big\| \leq \delta, \quad \big\|\Pi_0|\chi_s\rangle\big\| \leq \delta, \quad \text{and} \quad \sum_{t=1}^{s} \big\|U_t|\chi_{t-1}\rangle - |\chi_t\rangle\big\|^2 \leq \delta^2. \tag{10.91}$$

At this point it is helpful to take note of the telescoping sum

$$|\chi_t\rangle - U_t \cdots U_1 |\chi_0\rangle = \sum_{j=1}^{t} U_t \cdots U_{j+1} \big(|\chi_j\rangle - U_j |\chi_{j-1}\rangle\big), \tag{10.92}$$

which holds for each $t \in \{0, \ldots, s\}$. Here, $U_t \cdots U_{t+1}$ is to be interpreted as an empty product, equal to the identity matrix by definition. The sum telescopes because consecutive terms cancel, which does not depend on the matrices being unitary — a fact we will make use of in Chapter 11. Applying the triangle inequality and then the Cauchy–Schwarz inequality reveals that

$$\big\| |\chi_t\rangle - U_t \cdots U_1 |\chi_0\rangle \big\| \leq \sum_{j=1}^{t} \big\| |\chi_j\rangle - U_j |\chi_{j-1}\rangle \big\| \leq \sqrt{t}\, \delta \leq \sqrt{s}\, \delta. \tag{10.93}$$

In particular, we have

$$\big\| |\chi_t\rangle \big\| \leq \big\| |\chi_0\rangle \big\| + \sqrt{s}\, \delta, \tag{10.94}$$

again for each $t \in \{0, \ldots, s\}$.

Now let us prove an upper bound on $\| |\chi_0\rangle \|$, beginning with

$$\big\| |\chi_0\rangle \big\| = \big\| U |\chi_0\rangle \big\| \leq \big\| \Pi_0 U |\chi_0\rangle \big\| + \big\| \Pi_1 U |\chi_0\rangle \big\|. \tag{10.95}$$

For the first term, (10.93) with $t = s$ gives $\| U|\chi_0\rangle - |\chi_s\rangle \| \leq \sqrt{s}\, \delta$, which implies

$$\big\| \Pi_0 U |\chi_0\rangle - \Pi_0 |\chi_s\rangle \big\| \leq \sqrt{s}\, \delta \tag{10.96}$$

because projecting a vector cannot increase its norm. As $\| \Pi_0 |\chi_s\rangle \| \leq \delta$, it follows that

$$\big\| \Pi_0 U |\chi_0\rangle \big\| \leq \big(1 + \sqrt{s}\big)\delta. \tag{10.97}$$

For the second term, we split once again:

$$\big\| \Pi_1 U |\chi_0\rangle \big\| \leq \big\| \Pi_1 U \Delta_0 |\chi_0\rangle \big\| + \big\| \Pi_1 U \Delta_1 |\chi_0\rangle \big\|. \tag{10.98}$$

The vector $\Delta_0 |\chi_0\rangle$ has the form $|\xi\rangle \otimes |0^q\rangle$, so (10.88) gives

$$\big\| \Pi_1 U \Delta_0 |\chi_0\rangle \big\| \leq \frac{1}{2} \big\| \Delta_0 |\chi_0\rangle \big\| \leq \frac{1}{2} \big\| |\chi_0\rangle \big\|, \tag{10.99}$$

while (10.91) gives

$$\big\| \Pi_1 U \Delta_1 |\chi_0\rangle \big\| \leq \big\| \Delta_1 |\chi_0\rangle \big\| \leq \delta. \tag{10.100}$$

Altogether, we obtain

$$\big\| |\chi_0\rangle \big\| \leq \big(2 + \sqrt{s}\big)\delta + \frac{1}{2} \big\| |\chi_0\rangle \big\|, \tag{10.101}$$

and therefore

$$\big\| |\chi_0\rangle \big\| \leq 2\big(2+\sqrt{s}\big)\delta. \tag{10.102}$$

We may now combine this bound on $\big\| |\chi_0\rangle \big\|$ with (10.94) to obtain

$$\big\| |\chi_t\rangle \big\| \leq \big(4+3\sqrt{s}\big)\delta \leq 7\sqrt{s}\,\delta \tag{10.103}$$

for every $t \in \{0,\ldots,s\}$, and therefore

$$\big\| |\chi\rangle \big\|^2 = \sum_{t=0}^{s} \big\| |\chi_t\rangle \big\|^2 \leq 49s(s+1)\delta^2 \leq 49(s+1)^2\delta^2. \tag{10.104}$$

Thus, for $\delta = 1/(8(s+1))$, we conclude that every vector $|\chi\rangle \in \mathcal{L}$ satisfying (10.90) has $\| |\chi\rangle \|^2 \leq 49/64 < 1$. Consequently, if $|\chi\rangle \in \mathcal{L}$ is a unit vector, then at least one of the three inequalities in (10.90) fails. As each term of H is positive semidefinite, it follows that

$$\langle\chi|H|\chi\rangle \geq \langle\chi|H_{\text{in}}|\chi\rangle + \frac{1}{2}\sum_{t=1}^{s}\langle\chi|H_t|\chi\rangle + \langle\chi|H_{\text{out}}|\chi\rangle \geq \frac{1}{128(s+1)^2}. \tag{10.105}$$

By scaling, every vector $|\chi\rangle \in \mathcal{L}$ satisfies

$$\langle\chi|H|\chi\rangle \geq \frac{\| |\chi\rangle \|^2}{128(s+1)^2}. \tag{10.106}$$

Finally, let $|\chi\rangle$ be an arbitrary unit vector of $(\mathsf{C},\mathsf{W},\mathsf{Y})$. By (10.69), we have

$$\langle\chi|H|\chi\rangle = \langle\chi|\Pi_{\mathcal{L}}H\Pi_{\mathcal{L}}|\chi\rangle + \langle\chi|(\mathbb{1}-\Pi_{\mathcal{L}})H(\mathbb{1}-\Pi_{\mathcal{L}})|\chi\rangle. \tag{10.107}$$

The first term is at least $\|\Pi_{\mathcal{L}}|\chi\rangle\|^2/(128(s+1)^2)$ by (10.106). For the second, (10.55) along with the fact that the Hamiltonian terms are all positive semidefinite gives

$$\begin{aligned}\langle\chi|(\mathbb{1}-\Pi_{\mathcal{L}})H(\mathbb{1}-\Pi_{\mathcal{L}})|\chi\rangle &\geq \langle\chi|(\mathbb{1}-\Pi_{\mathcal{L}})H_{\text{clock}}(\mathbb{1}-\Pi_{\mathcal{L}})|\chi\rangle \\ &\geq \big\|(\mathbb{1}-\Pi_{\mathcal{L}})|\chi\rangle\big\|^2.\end{aligned} \tag{10.108}$$

As $\|\Pi_{\mathcal{L}}|\chi\rangle\|^2 + \|(\mathbb{1}-\Pi_{\mathcal{L}})|\chi\rangle\|^2 = 1$, adding the two bounds gives

$$\langle\chi|H|\chi\rangle \geq \frac{1}{128(s+1)^2}, \tag{10.109}$$

which establishes (10.89).

The reduction

We now have everything we need to prove the main result of the chapter.

Theorem 10.4 (Kitaev)**.** 5-LH *is* QMA*-complete.*

Proof. The containment 5-LH $\in$ QMA is the case $k = 5$ of Theorem 10.2. It remains to prove that 5-LH is QMA-hard. To this end, let $A = (A_{\text{yes}}, A_{\text{no}}) \in$ QMA, and let p, q, s, and $\{U_x\}$ be as described in Lemma 10.3. We will define an instance $f(x)$ of 5-LH for every $x \in \Sigma^*$, and then argue that the resulting function f is a Karp reduction from A to 5-LH. For this definition we take x to be implicit when writing H for the Hamiltonian corresponding to U_x, as in (10.47), which acts on $n = s + p + q$ qubits.

Each of the $2s + q$ terms of H is a projection that acts nontrivially on at most 5 qubits, as shown in Section 10.4. By (10.4), each such term can therefore be expressed as a real linear combination of at most $4^5 = 1024$ Pauli operations in $\mathcal{P}_n$, each having weight at most 5. Specifically, when each unitary U_t corresponds to a single gate in the gate set (10.36), each term of H can be expressed as a matrix having entries contained in the set $\{0, 1, \pm\frac{1}{4}, \pm\frac{1}{2}, \pm\frac{i}{2}\}$. By the 5-locality of each term, and considering the formula (10.4) for expressing a given matrix as a linear combination of Pauli operations, we may conclude that the coefficients needed to express each term of the Hamiltonian are integer multiples of $1/(4 \cdot 2^5) = 1/128$. These coefficients also lie in $[-1, 1]$, because each term has norm at most 1.

Listing all of the required Pauli operations and coefficients, for all of the terms of H, yields an expression

$$H = \gamma_1 P_1 + \cdots + \gamma_{m-1} P_{m-1} \tag{10.110}$$

where $m - 1 \leq 1024(2s + q)$, each $P_j \in \mathcal{P}_n$ has weight at most 5, and each γ_j is an integer multiple of $1/128$ in $[-1, 1]$.

Next, let $K = 512(s + 1)^2$ and define

$$P_m = \mathbb{1}^{\otimes n} \qquad \text{and} \qquad \gamma_m = -\frac{3}{K}, \tag{10.111}$$

noting that P_m has weight 0, so that

$$\gamma_1 P_1 + \cdots + \gamma_m P_m = H - \frac{3}{K}\mathbb{1}. \tag{10.112}$$

Finally, define the output of the reduction to be the instance

$$f(x) = \big(P_1, \ldots, P_m;\ \gamma_1, \ldots, \gamma_m;\ K\big) \tag{10.113}$$

of 5-LH, with each P_j expressed as a string in $\{\mathbb{1}, X, Y, Z\}^n$ and K written in unary. Note that $P_1, \ldots, P_m$ need not be distinct — this is not a requirement of the definition of k-LH.

The function f is polynomial-time computable: the family $\{U_x\}$ is polynomial-time uniform, each term of H is determined by at most one gate of U_x and the positions of the qubits it acts on, its Pauli coefficients are computed from its entries by (10.4), and the numbers n, m, and K are polynomially bounded.

If $x \in A_{\text{yes}}$, then by (10.37) some unit vector $|\psi\rangle$ satisfies

$$\Pr(U_x \text{ accepts } |\psi\rangle) \geq 1 - \frac{1}{256(s+1)}, \tag{10.114}$$

so (10.83) gives

$$\lambda_{\min}(H) \leq \frac{1}{256(s+1)^2} = \frac{2}{K}, \tag{10.115}$$

and therefore

$$\lambda_{\min}\left(H - \frac{3}{K}\mathbb{1}\right) \leq -\frac{1}{K}. \tag{10.116}$$

If $x \in A_{\text{no}}$, then by (10.38) every unit vector $|\psi\rangle$ satisfies $\Pr(U_x \text{ accepts } |\psi\rangle) \leq 1/4$, so (10.89) gives

$$\lambda_{\min}(H) \geq \frac{1}{128(s+1)^2} = \frac{4}{K}, \tag{10.117}$$

and therefore

$$\lambda_{\min}\left(H - \frac{3}{K}\mathbb{1}\right) \geq \frac{1}{K}. \tag{10.118}$$

Thus $A \leq_{\text{m}}^{\text{p}}$ 5-LH, and as $A \in$ QMA was arbitrary, 5-LH is QMA-hard. □

Chapter 11

Quantum advice

In this chapter we consider the notion of *quantum advice*. Similar to QMA, a quantum computation is provided with a quantum state, which influences whether the computation accepts or rejects a given input string. Unlike QMA, the quantum state is fully trusted in the setting of advice, though it is limited in a different way: the advice state may only depend upon the *length* of the input rather than the input itself.

The main result of the chapter is the containment expressed symbolically as $\mathrm{BQP/qpoly} \subseteq \mathrm{PP/poly}$. In words, polynomial-length quantum advice is no more useful to a polynomial-time quantum computation than polynomial-length classical advice is to a PP computation.

11.1 Classical advice

Quantum advice is a natural quantum extension of classical advice, which was first introduced by Richard Karp and Richard Lipton in 1980. The basic idea is that a trusted *advice string* is provided to a DTM, or other computational model, to aid it in its computation. As suggested above for the quantum setting, the advice is trusted but can only depend upon the input length n and not on which specific string of length n is input.

To formalize the notion of polynomial-length (classical) advice, we define the class $\mathcal{C}/\mathrm{poly}$ for an arbitrary complexity class $\mathcal{C}$. Here, the notation "/poly" indicates polynomial-length advice. Other advice classes, including $\mathcal{C}/\log$, $\mathcal{C}/\mathrm{const}$, and $\mathcal{C}/\exp$ (for logarithmic-, constant-, and exponential-length advice), can also be defined in an analogous way, but we will only discuss polynomial-length advice in this chapter.

Definition 11.1. Let $\mathcal{C}$ be a complexity class of promise problems. A promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ over an alphabet Σ is contained in $\mathcal{C}/\text{poly}$ if there exist a promise problem $B = (B_{\text{yes}}, B_{\text{no}}) \in \mathcal{C}$, a polynomial resource bound q, and a family of binary strings $\{w_n : n \in \mathbb{N}\}$ with $|w_n| \leq q(n)$ for every n, such that

$$x \in A_{\text{yes}} \implies \langle x, w_n \rangle \in B_{\text{yes}} \tag{11.1}$$

$$x \in A_{\text{no}} \implies \langle x, w_n \rangle \in B_{\text{no}} \tag{11.2}$$

for every $n \in \mathbb{N}$ and every string $x \in \Sigma^n$.

A few interesting facts about classical advice follow. The first fact is that a language L is in P/poly if and only if it is decided by polynomial-size Boolean circuits.

Proposition 11.2. *A language $L \subseteq \{0,1\}^*$ is contained in P/poly if and only if there exists a (not necessarily uniformly generated) Boolean circuit family $\{C_n : n \in \mathbb{N}\}$ that decides L and satisfies $\text{size}(C_n) = O(n^c)$ for some constant c.*

Proof sketch. If $L \in \text{P}/\text{poly}$, then there is a polynomial-time 1-DTM M and a set of polynomial-length advice strings $\{w_n : n \in \mathbb{N}\}$ such that M accepts $\langle x, w_{|x|} \rangle$ if and only if $x \in L$. By Theorem 3.11, which makes use of the tableau construction of Section 3.3, there is a polynomial-time uniform (and therefore polynomial-size) family $\{D_m : m \in \mathbb{N}\}$ of Boolean circuits for simulating M. A polynomial-size circuit C_n for $L \cap \{0,1\}^n$ is then obtained by hard-coding the appropriate advice string w_n into D_m, where $m = |\langle x, w_n \rangle|$ for every $x \in \{0,1\}^n$. That is,

$$C_n(x) = D_m(\langle x, w_n \rangle) \tag{11.3}$$

for every $n \in \mathbb{N}$ and $x \in \{0,1\}^n$.

On the other hand, if there exists a family $\{C_n : n \in \mathbb{N}\}$ of polynomial-size Boolean circuits for L, then we may choose advice strings that encode these circuits:

$$w_n = \langle C_n \rangle \tag{11.4}$$

for each $n \in \mathbb{N}$. As the circuit C_n given as advice on any input $x \in \{0,1\}^n$ can be evaluated on the input x in polynomial time, we conclude that $L \in \text{P}/\text{poly}$. □

The second interesting fact about classical advice is the following theorem, whose proof we have actually already covered, though it was not stated in terms of advice.

Theorem 11.3 (Adleman)**.** $\text{BPP} \subseteq \text{P}/\text{poly}$.

Proof. Let $A = (A_{\text{yes}}, A_{\text{no}})$ be a promise problem over an alphabet Σ contained in BPP. By Corollary 2.17, there exist a polynomial resource bound q, a language $C \in \text{P}$, and a family of strings $\{y_n : n \in \mathbb{N}\}$ with $|y_n| = q(n)$ for every $n \in \mathbb{N}$ such that $x \in A_{\text{yes}} \implies \langle x, y_n \rangle \in C$ and $x \in A_{\text{no}} \implies \langle x, y_n \rangle \notin C$. That $A \in \text{P/poly}$ therefore follows immediately by Definition 11.1. □

Finally, let us observe the following relevant but essentially trivial fact.

Proposition 11.4. $\text{PostBQP/poly} = \text{PP/poly}$.

This is immediate from $\text{PostBQP} = \text{PP}$ (Theorem 7.6) because Definition 11.1 only depends on the class $\mathcal{C}$ and not on a specific way of defining it.

11.2 Quantum advice

Now let us consider quantum advice, where classical advice strings are replaced by quantum states. We will only consider quantum advice states on a polynomial number of qubits, but (just like for classical advice) the notion can also be considered for quantum advice states of different sizes. We also only consider BQP computations.

The class BQP/qpoly

A definition of the complexity class BQP/qpoly follows. The notation "/qpoly" in this context refers to polynomial-length quantum advice.

Definition 11.5. A promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ over an alphabet Σ is contained in BQP/qpoly if there exist a polynomial resource bound p, a polynomial-time uniform family $\{Q_x : x \in \Sigma^*\}$ of quantum circuits, where each Q_x has p input qubits and one output qubit, and a family $\{|\psi_n\rangle : n \in \mathbb{N}\}$ of state vectors, where each $|\psi_n\rangle$ is a state vector of $p(n)$ qubits, such that

$$x \in A_{\text{yes}} \implies \Pr(Q_x \text{ accepts } |\psi_n\rangle) \geq \frac{2}{3} \tag{11.5}$$

$$x \in A_{\text{no}} \implies \Pr(Q_x \text{ accepts } |\psi_n\rangle) \leq \frac{1}{3} \tag{11.6}$$

for every $n \in \mathbb{N}$ and every string $x \in \Sigma^n$.

Thus, the set-up is similar to QMA in the sense that a polynomial-time uniform family of quantum circuits takes a quantum state as input and outputs a decision to accept or reject — but this time the quantum state is trusted and can only depend on the length of the input.

Remark 11.6. There is a natural alternative definition for BQP/qpoly that replaces the family $\{|\psi_n\rangle : n \in \mathbb{N}\}$ of state vectors with a family $\{\rho_n : n \in \mathbb{N}\}$ of mixed states on a polynomial number of qubits. The definitions are equivalent as any mixed quantum advice state ρ_n can be *purified* to a state on twice as many qubits, which leaves ρ_n when the additional qubits are traced out. Each quantum circuit Q_x may then simply ignore the second half of the qubits of the resulting pure advice states, effectively inputting the corresponding mixed advice state.

Error reduction

Next, we observe that error reduction works for BQP/qpoly, where each advice state for the error-reduced computation consists of many copies of the original advice state.

Proposition 11.7 (Error reduction for BQP/qpoly)**.** *Let $A = (A_{\text{yes}}, A_{\text{no}})$ be a promise problem contained in* BQP/qpoly*, taking p, $\{Q_x : x \in \Sigma^*\}$, and $\{|\psi_n\rangle : n \in \mathbb{N}\}$ as in Definition 11.5, and let r be a polynomial resource bound. There exists a polynomial-time uniform family $\{R_x : x \in \Sigma^*\}$ of quantum circuits, where each R_x has Np input qubits for $N = 18r$ and one output qubit, such that*

$$x \in A_{\text{yes}} \implies \Pr(R_x \text{ accepts } |\psi_n\rangle^{\otimes N}) \geq 1 - 2^{-r(n)} \tag{11.7}$$

$$x \in A_{\text{no}} \implies \Pr(R_x \text{ accepts } |\psi_n\rangle^{\otimes N}) \leq 2^{-r(n)} \tag{11.8}$$

for every $n \in \mathbb{N}$ and every string $x \in \Sigma^n$.

Proof. Let $\{R_x : x \in \Sigma^*\}$ be the family obtained by applying Construction 9.1 to each circuit Q_x, for $N = 18r$ and $t = N/2$. As in the proof of Theorem 9.2, this family is polynomial-time uniform: each R_x consists of N copies of Q_x followed by a threshold computation, all of which can be generated in polynomial time.

Now fix $n \in \mathbb{N}$ and a string $x \in \Sigma^n$, and suppose that R_x is run on the state $|\psi_n\rangle^{\otimes N}$. The N copies of Q_x each input a copy of $|\psi_n\rangle$, so the corresponding random variables $X_1, \ldots, X_N \in \{0, 1\}$ representing the outcomes of these computations are independent, each having expectation $\mu = \Pr(Q_x \text{ accepts } |\psi_n\rangle)$. An

application of Lemma 2.14 reveals the required implications: if $x \in A_{\text{yes}}$, then $\mu \geq 2/3$, so (2.31) gives

$$\Pr(R_x \text{ rejects } |\psi_n\rangle^{\otimes N}) = \Pr\big(X_1 + \cdots + X_N < N/2\big) \leq 2^{-r(n)}; \tag{11.9}$$

and if $x \in A_{\text{no}}$, then $\mu \leq 1/3$, and (2.32) gives

$$\Pr(R_x \text{ accepts } |\psi_n\rangle^{\otimes N}) = \Pr\big(X_1 + \cdots + X_N \geq N/2\big) \leq 2^{-r(n)}. \tag{11.10}$$

This completes the proof. □

Note that the in-place error reduction method for QMA cannot be applied in a straightforward way to obtain error reduction for BQP/qpoly. Consider why!

11.3 Nondestructive verification

The main result of the chapter, proved in Section 11.4, makes use of an interesting fact: the existence of a quantum state that passes a collection of verifications with very high probability allows one to learn a quantum state that also passes the same verifications, possibly with a somewhat lower probability. The procedure requires postselection, so it is not efficient in the sense that it might only succeed with very small probability. It is, however, remarkably efficient in that the learning itself only needs to involve a small number of the given collection of verifications — *linear in the number of qubits* being tested — even though the collection of verifications may be arbitrarily large. This section describes how this method works in isolation.

Single verifications

Consider the following definition, which amounts to step 1 together with a single iteration of step 2 of the in-place error reduction method described in Construction 9.3.

Definition 11.8. Let U be a unitary quantum circuit acting on $p + q$ qubits. The *nondestructive verification* defined by U is the following procedure, which is applied to a p-qubit register W.

1. Initialize a q-qubit register Y to the all-zero state.
2. Apply U to the pair of registers (W, Y).
3. Perform a standard basis measurement on the first qubit of W.

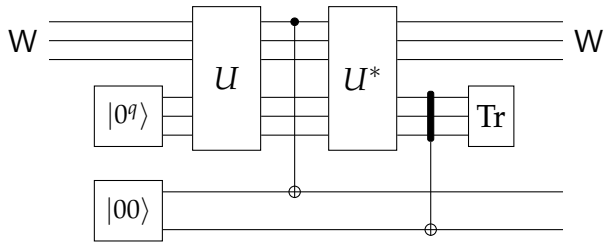


Figure 11.1: A nondestructive verification, with its measurements implemented by controlled-NOT gates onto fresh qubits. The tall oval is an OR-control, which does nothing when every control qubit is set to 0 (in the standard basis) and otherwise applies the NOT operation to the target. The verification passes when the first measurement outcome is 1 and the second is 0.

4. Apply U^* to (W, Y).

5. Perform a standard basis measurement on every qubit of Y.

The verification *passes* if the measurement in step 3 gives the outcome 1 and the measurement in step 5 gives the outcome 0^q. The register W is output and Y is discarded.

Remark 11.9. Step 5 may alternatively perform the measurement described by the projections $|0^q\rangle\langle 0^q|$ and $\mathbb{1}^{\otimes q} - |0^q\rangle\langle 0^q|$, similar to what is done for in-place error reduction for QMA. For the purposes of this discussion the two options are equivalent, as we will only be concerned with what happens when the measurement outcome corresponds to the projection $|0^q\rangle\langle 0^q|$.

Our focus will be on the probability that such a verification passes, along with a bound on how much the initial state of W is disturbed in this event. (Unlike in-place error reduction for QMA, we will not need to consider what happens for other measurement outcomes.)

First, supposing that W is initially in the pure state $|\psi\rangle$, the probability that the measurement in step 3 gives the outcome 1 is

$$\begin{aligned} \Pr(U \text{ accepts } |\psi\rangle) &= \big\| \big(\langle 1| \otimes \mathbb{1}^{\otimes(p+q-1)}\big) U \big(|\psi\rangle \otimes |0^q\rangle\big) \big\|^2 && (11.11) \\ &= \big\| M|\psi\rangle \big\|^2 && (11.12) \\ &= \langle \psi | M^* M | \psi \rangle && (11.13) \end{aligned}$$

for

$$M = \big(|1\rangle\langle 1| \otimes \mathbb{1}^{\otimes(p+q-1)}\big) U \big(\mathbb{1}^{\otimes p} \otimes |0^q\rangle\big). \tag{11.14}$$

Defining $P = M^*M$, we therefore have

$$\Pr(U \text{ accepts } |\psi\rangle) = \langle\psi|P|\psi\rangle. \tag{11.15}$$

This is the same calculation as in Section 9.2: viewing U as a unitary verification circuit for QMA, with W holding the witness and Y the ancillary qubits, the quantity $\Pr(U \text{ accepts } |\psi\rangle)$ is the probability that the witness $|\psi\rangle$ is accepted. This matrix P satisfies $0 \leq P \leq \mathbb{1}^{\otimes p}$, and represents the measurement operator (or POVM element) that corresponds to just steps 1–3 of the verification procedure leading to the outcome 1. In the case that W starts in a mixed state ρ, the probability to obtain the outcome 1 is equal to $\mathrm{Tr}(P\rho)$.

To understand what happens when the entire verification passes, so that the measurement outcomes in steps 3 and 5 are 1 and 0^q, respectively, we first observe that

$$\big(\mathbb{1}^{\otimes p} \otimes \langle 0^q|\big)U^*\big(|1\rangle\langle 1| \otimes \mathbb{1}^{\otimes(p+q-1)}\big)U\big(|\psi\rangle \otimes |0^q\rangle\big) = M^*M|\psi\rangle = P|\psi\rangle. \tag{11.16}$$

This represents the unnormalized state of W produced by these measurement outcomes. The probability that the verification passes is therefore

$$\big\|P|\psi\rangle\big\|^2 = \langle\psi|P^2|\psi\rangle, \tag{11.17}$$

while the state of W conditioned on the verification passing is obtained by normalizing:

$$\frac{P|\psi\rangle}{\|P|\psi\rangle\|}. \tag{11.18}$$

By linearity, if W is initially in a state ρ, the verification passes with probability $\mathrm{Tr}(P\rho P)$, in which case the output state of W is

$$\frac{P\rho P}{\mathrm{Tr}(P\rho P)}. \tag{11.19}$$

Finally, we note the following fact. Under the assumption that $|\psi\rangle$ is a unit vector such that

$$\langle\psi|P|\psi\rangle \geq 1 - \varepsilon \tag{11.20}$$

for $\varepsilon \geq 0$, we have

$$\big\||\psi\rangle - P|\psi\rangle\big\|^2 = \langle\psi|(\mathbb{1} - P)^2|\psi\rangle \leq \langle\psi|(\mathbb{1} - P)|\psi\rangle \leq \varepsilon \tag{11.21}$$

and therefore

$$\big\||\psi\rangle - P|\psi\rangle\big\| \leq \sqrt{\varepsilon}. \tag{11.22}$$

Note that the first inequality in (11.21) follows from the fact that every eigenvalue of $\mathbb{1} - P$ is contained in the interval $[0, 1]$ and therefore cannot grow when squared, giving $(\mathbb{1} - P)^2 \leq \mathbb{1} - P$.

Sequences of verifications

Now suppose multiple nondestructive verifications are given, defined by unitary operations $U_1, \ldots, U_T$ on $p + q$ qubits. We will consider what happens when these nondestructive verifications are performed in sequence, starting with the one defined by U_1.

Let $P_1, \ldots, P_T$ be the matrices described as P above when U is replaced by $U_1, \ldots, U_T$, respectively. Assuming W begins in the state $|\psi\rangle$, the vector

$$P_T \cdots P_1 |\psi\rangle \tag{11.23}$$

describes the event that every one of the nondestructive verifications passes. That is,

$$\big\| P_T \cdots P_1 |\psi\rangle \big\|^2 \tag{11.24}$$

is the probability of this event, and

$$\frac{P_T \cdots P_1 |\psi\rangle}{\| P_T \cdots P_1 |\psi\rangle \|} \tag{11.25}$$

is the state of W conditioned on this event. More generally, if W is initially in a state described by a density matrix ρ, then every nondestructive verification passes with probability $\mathrm{Tr}(P_T \cdots P_1 \rho P_1 \cdots P_T)$, and conditioned on this event the state of W becomes

$$\frac{P_T \cdots P_1 \rho P_1 \cdots P_T}{\mathrm{Tr}(P_T \cdots P_1 \rho P_1 \cdots P_T)}. \tag{11.26}$$

Now suppose that $|\psi\rangle$ is a quantum state for which every one of the verifications defined by $U_1, \ldots, U_T$ would give the outcome 1 (in step 3 of Definition 11.8) with probability at least $1 - \varepsilon$, when performed alone:

$$\Pr(U_t \text{ accepts } |\psi\rangle) = \langle \psi | P_t | \psi \rangle \geq 1 - \varepsilon \tag{11.27}$$

for every $t \in \{1, \ldots, T\}$. We will conclude that

$$\big\| |\psi\rangle - P_T \cdots P_1 |\psi\rangle \big\| \leq T\sqrt{\varepsilon}. \tag{11.28}$$

To see this, observe that the telescoping sum (10.92) from Chapter 10, which holds for arbitrary matrices, gives

$$|\psi\rangle - P_T \cdots P_1 |\psi\rangle = \sum_{t=1}^{T} P_T \cdots P_{t+1} (\mathbb{1} - P_t) |\psi\rangle \tag{11.29}$$

when each vector $|\chi_j\rangle$ is replaced by $|\psi\rangle$ and each matrix U_j by P_j. Each P_t has norm at most 1, so the triangle inequality together with (11.22) gives

$$\big\| |\psi\rangle - P_T \cdots P_1 |\psi\rangle \big\| \leq \sum_{t=1}^{T} \big\| (\mathbb{1} - P_t)|\psi\rangle \big\| \leq T\sqrt{\varepsilon}. \tag{11.30}$$

Observing that (11.28) implies

$$\|P_T \cdots P_1 |\psi\rangle\| \geq 1 - T\sqrt{\varepsilon}, \tag{11.31}$$

we have that if $T \leq 1/\sqrt{\varepsilon}$, then $|\psi\rangle$ passes all T nondestructive verifications with probability

$$\big\|P_T \cdots P_1 |\psi\rangle\big\|^2 \geq \big(1 - T\sqrt{\varepsilon}\big)^2. \tag{11.32}$$

Learning through postselection

Finally, we are prepared to discuss the learning procedure suggested at the start of the section.

We now assume that an *arbitrary* collection $\mathcal{V}$ of nondestructive verifications on a p-qubit register is given, each represented by a positive semidefinite matrix P with $0 \leq P \leq \mathbb{1}^{\otimes p}$ as described above. In particular, the set $\mathcal{V}$ can be exponential in size, or even infinite. We also assume that there exists a p-qubit quantum state $|\psi\rangle$ such that

$$\langle\psi|P|\psi\rangle \geq 1 - \frac{1}{64(p+1)^2} \tag{11.33}$$

for every $P \in \mathcal{V}$, though neither this state nor a description of it is made available. The goal is to create, through postselection, a p-qubit quantum state ρ such that

$$\mathrm{Tr}(P\rho) \geq \frac{2}{3} \tag{11.34}$$

for every $P \in \mathcal{V}$.

The strategy begins with W being in a completely mixed state of p qubits, which we will denote as

$$\rho_0 = 2^{-p}\mathbb{1}^{\otimes p}. \tag{11.35}$$

We set $t \leftarrow 0$ and iterate:

1. If the current state ρ_t of W satisfies $\mathrm{Tr}(P\rho_t) \geq 2/3$ for every $P \in \mathcal{V}$, then the goal is accomplished: stop and output W. Otherwise, choose $P_{t+1} \in \mathcal{V}$ for which $\mathrm{Tr}(P_{t+1}\rho_t) < 2/3$.

2. Perform the nondestructive verification P_{t+1} on W, and postselect on its passing, which produces the state ρ_{t+1}.

3. Set $t \leftarrow t+1$. If t exceeds $2(p+1)$, then the strategy fails; otherwise, return to step 1.

We will prove that this procedure never actually fails: it necessarily terminates when $t = T$ for some choice of $T \leq 2p+1$.

Before continuing further, it is appropriate to remark that executing this procedure requires making valid choices for each P_t. We do not assume a feasible way to make these choices, but take as given that valid choices are made. This is where classical advice is required in the next section.

Based on the analysis of the previous two subsections, we may express each state ρ_t, representing the state of W after t iterations of the learning strategy, as follows:

$$\rho_t = \frac{P_t \cdots P_1 \rho_0 P_1 \cdots P_t}{\operatorname{Tr}(P_t \cdots P_1 \rho_0 P_1 \cdots P_t)} = \frac{P_t \cdots P_1 P_1 \cdots P_t}{\operatorname{Tr}(P_t \cdots P_1 P_1 \cdots P_t)}. \tag{11.36}$$

Now let us verify that the strategy must stop when $t = T$ for some choice of $T \leq 2p+1$. First, suppose that P_t has been chosen, for some t satisfying $1 \leq t \leq 2(p+1)$. We have the following, where the inequality uses the fact that $P_t^2 \leq P_t$; just like $(\mathbb{1} - P)^2 \leq \mathbb{1} - P$ above, every eigenvalue of P_t is contained in the interval $[0,1]$ and therefore cannot grow when squared.

$$\begin{aligned}
\operatorname{Tr}(P_t \cdots P_1 P_1 \cdots P_t) &= \operatorname{Tr}(P_t^2 P_{t-1} \cdots P_1 P_1 \cdots P_{t-1}) && (11.37)\\
&\leq \operatorname{Tr}(P_t P_{t-1} \cdots P_1 P_1 \cdots P_{t-1}) && (11.38)\\
&= \operatorname{Tr}(P_t \rho_{t-1}) \cdot \operatorname{Tr}(P_{t-1} \cdots P_1 P_1 \cdots P_{t-1}) && (11.39)\\
&< \frac{2}{3} \cdot \operatorname{Tr}(P_{t-1} \cdots P_1 P_1 \cdots P_{t-1}) && (11.40)
\end{aligned}$$

The same inequality may be applied iteratively for decreasing values of t, and we have $\operatorname{Tr}(\mathbb{1}^{\otimes p}) = 2^p$, which implies

$$\operatorname{Tr}(P_t \cdots P_1 P_1 \cdots P_t) < \left(\frac{2}{3}\right)^t \cdot 2^p. \tag{11.41}$$

Bringing the state $|\psi\rangle$ satisfying (11.33) into the analysis, we have $|\psi\rangle\langle\psi| \leq \mathbb{1}^{\otimes p}$ and therefore

$$\operatorname{Tr}(P_t \cdots P_1 P_1 \cdots P_t) \geq \operatorname{Tr}(P_t \cdots P_1 |\psi\rangle\langle\psi| P_1 \cdots P_t) = \| P_t \cdots P_1 |\psi\rangle \|^2. \tag{11.42}$$

Finally, as $t \leq 2(p+1)$, we conclude by (11.32) that

$$\|P_t \cdots P_1 |\psi\rangle\|^2 \geq \left(1 - \frac{t}{8(p+1)}\right)^2 \geq \left(\frac{3}{4}\right)^2 \geq \frac{1}{2}. \tag{11.43}$$

Consequently,

$$\frac{1}{2} < \left(\frac{2}{3}\right)^t \cdot 2^p \leq 2^{p - t/2}, \tag{11.44}$$

which implies that $t < 2(p+1)$. The matrix $P_{2(p+1)}$ is therefore never chosen, so the strategy never fails: it stops when $t = T$ for some $T \leq 2p+1$.

By the inequality (11.43) for $t = T$, we conclude that with probability

$$\mathrm{Tr}(P_T \cdots P_1 \rho_0 P_1 \cdots P_T) = 2^{-p} \,\mathrm{Tr}(P_T \cdots P_1 P_1 \cdots P_T) \geq 2^{-(p+1)}, \tag{11.45}$$

all T postselections along the way succeed, resulting in a final state ρ_T of W satisfying $\mathrm{Tr}(P\rho_T) \geq 2/3$ for every $P \in \mathcal{V}$.

11.4 BQP/qpoly is contained in PP/poly

Now we are prepared to prove the main result of the chapter.

Theorem 11.10 (Aaronson)**.** BQP/qpoly $\subseteq$ PP/poly.

Proof. Let $A = (A_{\mathrm{yes}}, A_{\mathrm{no}})$ be a promise problem over an alphabet Σ contained in BQP/qpoly. For each $x \in A_{\mathrm{yes}} \cup A_{\mathrm{no}}$, let $a(x) = 1$ if $x \in A_{\mathrm{yes}}$ and $a(x) = 0$ if $x \in A_{\mathrm{no}}$. That is, $a(x)$ represents the correct answer for A on each input x satisfying the promise.

The first step is to perform error reduction. By Proposition 11.7, we may conclude that there exists a collection of states

$$\{|\psi_n\rangle \,:\, n \in \mathbb{N}\}, \tag{11.46}$$

where each $|\psi_n\rangle$ is a pure state of $p(n)$ qubits for a polynomial resource bound p, such that the following is true. For every polynomial resource bound r, and setting $N = 18r$, there exists a polynomial-time uniform family $\{R_x \,:\, x \in \Sigma^*\}$ of quantum circuits, where each R_x takes Np qubits as input and produces one output qubit, such that

$$x \in A_{\mathrm{yes}} \implies \Pr(R_x \text{ accepts } |\psi_{|x|}\rangle^{\otimes N}) \geq 1 - 2^{-r} \tag{11.47}$$

$$x \in A_{\mathrm{no}} \implies \Pr(R_x \text{ accepts } |\psi_{|x|}\rangle^{\otimes N}) \leq 2^{-r} \tag{11.48}$$

for every string $x \in \Sigma^*$. Taking $r = p + 64$ guarantees that

$$2^{-r} \leq \frac{1}{64(Np+1)^2}, \tag{11.49}$$

as a routine calculation reveals.

Next, let $\{U_x : x \in \Sigma^*\}$ be a polynomial-time uniform family of unitary quantum circuits so that each U_x dilates R_x, as in Proposition 5.2. Thus, each U_x acts on $Np + q$ qubits, for a polynomial resource bound q, with the first qubit corresponding to the output qubit of R_x. For each $x \in A_{\text{yes}} \cup A_{\text{no}}$, define

$$V_x = \bigl(X^{1-a(x)} \otimes \mathbb{1}^{\otimes(Np+q-1)}\bigr) U_x, \tag{11.50}$$

which is U_x followed by a NOT gate on the output qubit when $a(x) = 0$. For a given input length $n \in \mathbb{N}$, the circuits $\{V_x\}$ define a set of nondestructive verifications from which a suitable substitute for the advice state $|\psi_n\rangle^{\otimes N}$ will be learned by postselection. Specifically, letting P_x be the matrix associated with V_x as in Section 11.3, consider the set

$$\mathcal{V}_n = \bigl\{P_x : x \in \Sigma^n \cap (A_{\text{yes}} \cup A_{\text{no}})\bigr\}. \tag{11.51}$$

For each $P_x \in \mathcal{V}_n$, we have

$$\operatorname{Tr}(P_x \rho) = \langle a(x)|R_x(\rho)|a(x)\rangle \tag{11.52}$$

for every Np-qubit state ρ, by the calculation in Section 11.3: step 3 of the verification defined by V_x gives the outcome 1 exactly when a measurement of the output qubit of U_x would give $a(x)$.

By the analysis of the previous section, together with the existence of the collection (11.46) satisfying (11.47) and (11.48), and by (11.49) and (11.52), we may conclude the following for every fixed input length $n \in \mathbb{N}$. There must exist $T \leq 2Np + 1$ strings $y_1, \ldots, y_T \in \Sigma^n \cap (A_{\text{yes}} \cup A_{\text{no}})$ such that the state

$$\rho_T = \frac{P_{y_T} \cdots P_{y_1} P_{y_1} \cdots P_{y_T}}{\operatorname{Tr}(P_{y_T} \cdots P_{y_1} P_{y_1} \cdots P_{y_T})} \tag{11.53}$$

satisfies

$$\langle a(x)|R_x(\rho_T)|a(x)\rangle = \operatorname{Tr}(P_x \rho_T) \geq \frac{2}{3} \tag{11.54}$$

for every $x \in \Sigma^n \cap (A_{\text{yes}} \cup A_{\text{no}})$. Moreover, by (11.45), if the nondestructive verifications defined by $V_{y_1}, \ldots, V_{y_T}$ are performed in sequence on the completely mixed state of Np qubits, then all of them pass with probability at least $2^{-(Np+1)}$, leaving

the state ρ_T. The strings $y_1, \ldots, y_T$ need not be unique, but a definite choice can be made by taking each y_t to be the lexicographically first string that is a valid choice in the learning procedure.

The final step is to define a postselection computation that takes as input a string $x \in \Sigma^n$ together with the classical advice string

$$w_n = \langle y_1, a(y_1), y_2, a(y_2), \ldots, y_T, a(y_T) \rangle, \tag{11.55}$$

creates the state ρ_T using postselection, and then runs R_x on it to determine its final answer. It operates as follows.

1. Prepare an Np-qubit register W in the completely mixed state, using Np copies of the random-bit generator of Figure 5.1.

2. For $t = 1, \ldots, T$, perform the nondestructive verification defined by V_{y_t} on W. (The circuit V_{y_t} is determined by y_t and $a(y_t)$, both of which are given by the advice string.)

3. Set the postselection qubit to $|1\rangle$ if all T verifications passed, and to $|0\rangle$ otherwise.

4. Run R_x on W, and take its output qubit as the output qubit of the computation.

By the previous paragraph, the postselection succeeds with probability at least $2^{-(Np+1)} > 0$, and conditioned on its success, the output is $a(x)$ with probability $\langle a(x)|R_x(\rho_T)|a(x)\rangle \geq 2/3$, by (11.54). This implies $A \in \text{PostBQP/poly}$, and therefore $A \in \text{PP/poly}$ by Proposition 11.4. □